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Two neuropeptides that promote blood feeding in *Anopheles stephensi* mosquitoes

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eLife Assessment

This is a **valuable** study that integrates behavioral and molecular approaches to identify neuromodulators influencing blood-feeding behavior in the disease vector Anopheles stephensi. Through gene expression analyses across blood-seeking life stages and RNA interference experiments, the authors present **solid** evidence that co-knockdown of the neuromodulators short Neuropeptide F and RYamide affects blood-seeking states in A. stephensi. However, evidence demonstrating that these neuropeptides are sufficient to promote host-seeking is lacking.

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Abstract

Animals routinely need to make decisions about what to eat and when. These decisions are influenced not only by the availability and quality of food but also by the internal state of the animal, which needs to compute and give weights to these different variables before making a choice. Feeding preferences of female mosquitoes exemplify this behavioural plasticity. Both male and female mosquitoes usually feed on carbohydrate-rich sources of nectar or sap, but the female also feeds on blood, which is essential for egg development. This blood-appetite is modulated across the female's reproductive cycle, yet little is known about the factors that bring it about. We show that mated, but not virgin *Anopheles stephensi* females, a major vector of urban malaria in the Indian sub-continent and West Africa, suppress blood feeding between a blood meal and oviposition. We identify several candidate genes through transcriptomics of blood-deprived and -sated *An. stephensi* central brains that could modulate this behaviour. We show that *short neuropeptide F (sNPF)* and *RYamide (RYa)* act together to promote blood feeding and identify a cluster of cells in the subesophageal zone that expresses *sNPF* transcripts only in the blood-hungry state. Such females also have more *sNPF* transcripts in their midguts. Based on these data, we propose a model where increased *sNPF* levels in the brain and gut promote a state of blood-hunger, which drives feeding behaviour either by *sNPF's* action in the two tissues independently or via a communication between them. This occurs in the context of the action of *RYa* in the brain.

Introduction and Results

An animal's dietary choice is an outcome of its internal physiological state and external stimuli. Its changing nutritional demands, reproductive state, and developmental history are integrated with external cues before a food choice is made. For example, when deprived of a particular nutrient, like proteins, animals show a specific increase in protein-appetite (1–3). Mating influences food choices of female flies by increasing their preference for proteins, sugars, and salt for egg development and oviposition (4–8).

Changes in both sensory perception and central brain circuits have been shown to underlie such behavioural changes. For example, sensory neurons of starved animals become more sensitive to attractive cues and less sensitive to aversive ones, both in vertebrates and invertebrates (9–16). These changes are often brought about by neuromodulators and neuropeptides (9, 10, 17–22). In the central brain, internal nutrient or hormone sensors can monitor the animal's nutritional state. In mice, neurons in the hypothalamus express receptors for hunger and satiety signals like ghrelin and leptin (16, 23, 24). In insects, several neurons in the mushroom body, central complex, and subesophageal zone (SEZ) have been shown to integrate hunger and satiety signals to regulate feeding on sugars or proteins, choose between exploration or exploitation of food sources, or initiate or terminate feeding (25–33).

Female mosquitoes offer a striking model for dissecting how internal state and external cues shape dietary choices. Many female mosquitoes have a dual appetite for carbohydrate-rich nectar or sap – to meet energetic requirements – and protein-rich blood – for egg development. Which meal a female chooses to take is influenced by nutritional demands, reproductive state, and developmental history. For example, mosquitoes with low teneral reserves will take multiple blood meals for egg development (34–37). Nutritional state also influences other internal state outputs on blood feeding. For example, whether virgin females take blood meals or not is dependent on prior sugar feeding in most *Aedes* strains (38) (and references within).

Despite these species- and context-dependent variations, a consistent finding is that female mosquitoes oscillate between states of heightened and suppressed host seeking across successive reproductive cycles (figure 1A). For example, in *Ae. aegypti* and *An. gambiae*, newly emerged mosquitoes display no or low interest in the host, which increases as they age (34, 39–41) and once fed to repletion, host attraction is suppressed. The strength and timing of this suppression varies between species — being robust and well-characterised in mated *Ae. aegypti* (42–44), and more flexible in anophelines (36, 45, 46). Sugar feeding is not modulated in this way, in some cases it is preferred soon after emergence (40, 47).

The modulation of blood feeding provides a useful framework for understanding how internal and external cues shape meal choice. What underlying molecular cues represent internal physiological states to influence dietary choice is of interest not only because of its importance in public health, but also because such understanding has implications for other organisms. In *Ae. albopictus* anticipatory expression of the yolk protein precursor gene *vitellogenin 2* in the fat body promotes host-seeking behaviour (48). In *Ae. aegypti*, neuropeptide Y-like receptor 7 (NPYLR7), and more recently, short neuropeptide F (sNPF) and neuropeptide RYamide (RYa) have been shown to suppress host attraction, while neuropeptide F (NPF) promotes it (44, 49–52). Factors from the host can also both promote (ATP) (53) and terminate feeding (Fibrinopeptide A)(54).

Here we describe the blood- and sugar-feeding patterns of *Anopheles stephensi*, which is endemic to the Indian sub-continent but has expanded its range into West Africa, putting millions more people at risk of malaria (55). Neurotranscriptomics across the different conditions of blood-deprivation and -satiety allowed us to shortlist several candidate molecules that might promote blood-feeding behaviour. Through dsRNA-mediated knockdown of nine of them, we identified two neuropeptide genes – *sNPF* and *RYa* – that together promote blood feeding through their action in the brain and possibly the abdomen. We find that *sNPF* expression pattern reflects this requirement: a cluster of cells in the SEZ expresses *sNPF* transcripts only in the blood-hungry females and such females also have more *sNPF* transcripts in their midguts. *RYa* is not modulated similarly. Both receptors are expressed in the brain and the *sNPF receptor* is also expressed in the midgut. Together, these data are consistent with a model where an increase in levels of *sNPF* in the brain and the midgut promotes a state of blood-hunger in *An. stephensi*. This occurs in the context of *RYa* signalling in the brain.

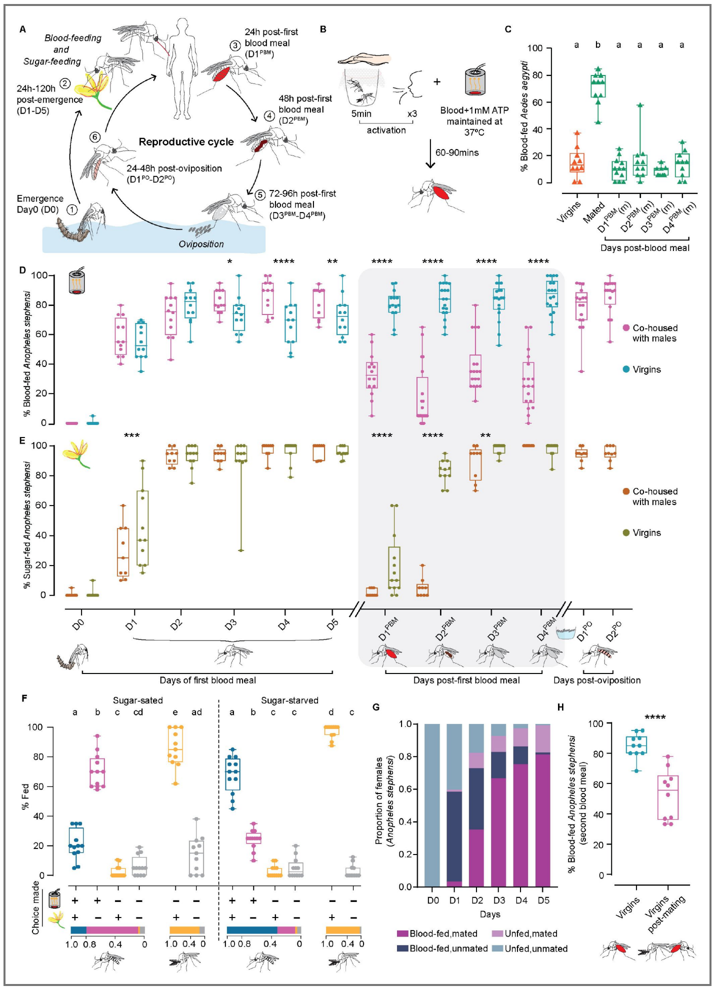


Fig. 1. Feeding behaviours of *An. stephensi*.

A. Reproductive cycle of *An. stephensi*. Upon emergence [1, D0-D5] females have a dual appetite for protein-rich blood and carbohydrate-rich sources of sugar [2]. After a blood meal [3,4,5, D1^{PBM}-D4^{PBM}], the eggs mature and are laid in water [6]. The next reproductive cycle can now begin. Striped abdomens indicate mated females. **B**. Blood-feeding assay. 0-7h-old mosquitoes were collected in cups and aged appropriately on sugar. Prior to the test, mosquitoes were 'activated' for 5 minutes by presenting a human hand, followed by 3 exhalations. Blood meals were presented through a Hemotek perfumed with human skin odours. Blood-fed mosquitoes were visually scored. **C**. Blood-feeding behaviour of *Ae. aegypti*. 8 days-old virgin and mated females were assessed for first blood meal. Fully fed mated females were assessed for second blood meal (D1^{PBM}-D4^{PBM}). 18-21 females/trial, n=10-12 trials/group. Kruskal-Wallis with Dunn's multiple comparisons test, $p<0.05$. Data labelled with different letters are significantly different. (m): mated. **D**. Blood-feeding behaviour of *An. stephensi*. Co-housed and virgin females were tested on the day of emergence (D0) and each day for the next 5 days (D1-D5). D1^{PBM}-D4^{PBM} represents blood feeding 1-4 days post-first blood meal and D1^{PO}-D2^{PO} represents blood feeding 1-2 days post-oviposition. 18-21 females/trial, n=9-20 trials/group. Generalized Linear Mixed Model with post-hoc pairwise comparisons using estimated marginal means and Bonferroni-corrected p-values; *$p<0.05$; **$p<0.01$; ****$p<0.0001$. **E**. Sugar-feeding behaviour of *An. stephensi*. Females of similar conditions as in (D) were tested for sugar feeding. 18-21 females/trial, n=9-20 trials/group. Generalized Linear Mixed Model with post-hoc pairwise comparisons using estimated marginal means and Bonferroni-corrected p-values; **$p<0.01$; ***$p<0.001$; ****$p<0.0001$. **F**. Choice assay for blood and sugar. Sugar-sated or sugar-starved females, co-housed with males (indicated by striped abdomens) were given a choice between blood and sugar and assessed for the choice made (bottom): blood and sugar (blue); blood only (magenta); sugar only (orange); none (grey). Co-housed males were assessed for sugar feeding only. Proportion of mosquitoes fed on each particular choice are shown at the bottom. 17-24 mosquitoes/trial; n=11-12 trials/condition. One-way ANOVA with Holm-Šídák multiple comparisons test, $p<0.05$. Data labelled with different letters are significantly different. **G**. Mating and blood feeding. Mating status of D0-D5 co-housed females assayed in (D) were determined post-hoc via spermatheca dissection. n=232-239 females analysed for each time point. **H**. Mating after blood feeding. Blood-fed virgins were allowed to mate and assayed for second blood meal. Age-matched virgins were used as controls. 15-20 females/trial, n=10 trials. Unpaired t-test; ****$p<0.0001$.

Blood feeding, but not sugar feeding, is modulated across the reproductive cycle

To assess the female *An. stephensi* appetite for blood through the reproductive cycle we designed a behavioural assay (figure 1B, S1A; see methods for details), and tested it on *Ae. aegypti*, a well-studied model. We were able to reproduce the previously reported results that mating enhances blood feeding in *Aedes* (56–58) and that once fed to repletion, females actively suppress blood feeding until oviposition (59) (figure 1C). We used this assay on *An. stephensi*, and employed Generalised Linear Mixed Model (GLMM) to compare appetites across conditions (see supplementary sheet 1). We found that these females were uninterested in blood when they emerge but that their appetite increased with age (magenta, D0-D5 in figure 1D, supplementary sheet 1:model 1). Once fed to repletion, like *Ae. aegypti, An. stephensi* also suppress blood feeding for at least four days after a blood meal (magenta, $D1^{PBM}$-$D4^{PBM}$ in figure 1D) and until oviposition, after which, the suppression was lifted to pre-blood meal levels (magenta, $D1^{PO}$-$D2^{PO}$ in figure 1D). Unlike blood feeding, sugar feeding was not modulated similarly barring a brief suppression post-blood meal, which was not dependent on oviposition (figure 1E; methods and S1G for details, supplementary sheet 1:model 4, 5, 6).

To ascertain whether sugar and blood represent parallel appetites independent of each other, we presented sugar-sated co-housed females a choice between the two. Under these conditions, 70% of the females chose to feed on blood alone and only 21% took both meals (figure 1F). When co-housed females were deprived of sugars for a day prior to testing, the pattern reversed: 68% of the females now took both sugar and blood meals and only 24% fed on blood alone (figure 1F). As expected, males fed robustly on sugar in both condi-tions (grey, figure 1F).

Since females took blood meals regardless of their prior sugar feeding status and only sugar feeding was selectively suppressed by prior sugar access, we infer that these two represent parallel appetites regulated independently. The fact that most females in the starved group took both sugar- and blood meals, while most females in the sated condition ignored sugar, supports this interpretation. An alternate interpretation of these data are that two represent hierarchical appetites: that some sugar feeding is necessary to stimulate blood feeding. Determining the order of feeding events in the choice assay would be necessary to comment on this. However, prior reports that females can survive on blood meals alone (with no access to plant sugars (36)) argue against this model. Together, these data suggest that in newly emerged females, blood and sugar represent parallel appetites. We note that their interdependence may emerge later in the reproductive cycle ($D1^{PBM}$-$D2^{PBM}$ in figure 1E).

As mating promotes the first blood meal in *Ae. aegypti*, we tested if this was true for *An. stephensi*. A post-hoc dissection of each female's spermatheca (where sperm is stored after mating) revealed that many of the blood-fed females in the above assay were unmated (figure 1G). To confirm this, we assayed blood-feeding behaviour in virgin females and found that virgin *An. stephensi*, unlike *Ae. aegypti*, have a robust appetite for blood and that mating enhances it only marginally (cyan, D0-D5, figure 1D, supplementary sheet 1:model 1, 2). Interestingly, however, these virgins, unlike mated *An. stephensi* (or *Ae. aegypti)* females, failed to suppress their blood-appetite, even after engorging on blood. Instead, they continued to blood-feed for up to four days after their first blood meal (cyan, $D1^{PBM}$ - $D4^{PBM}$ in figure 1D, supplementary sheet 1:model 3). A flipped order of the two behaviours - virgins were first given a blood meal then allowed to mate – also resulted in suppression of the appetite (figure 1H). This suggests that the post-blood meal suppression of blood-appetite in *An. stephensi* is dependent upon mating status.

Together, these data describe female *An. stephensi's* feeding behaviours: 1) in newly emerged females blood and sugar represent parallel appetites, 2) blood-, but not sugar-appetite (barring a brief effect), is suppressed between a blood meal and oviposition, and 3) mating is not necessary for blood feeding in newly emerged females, but it is necessary for its suppression between a blood meal and oviposition.

Host seeking is modulated across the reproductive cycle

The modulation of blood feeding through the reproductive cycle could be due to 1) modulation in peripheral sensitivity to host kairomones, 2) modulation by central brain circuits, or 3) modulation in both peripheral and central circuit elements. Extensive data in other mosquito species suggest that sensitivity to host kairomones change as females age (39, 60–63) and following a blood meal (46, 64, 65). We tested this in *An. stephensi*.

Because of the multi-sensory nature of host seeking, we used a two-choice behavioural assay where multiple host cues were provided in one arm while the other had none. Mosquitoes had to make a choice between the two arms (figure 2A). When no host cues are provided in either (blank) even blood-hungry females distribute equally between the two arms demonstrating the assay has no bias (black, figure 2B). Using this assay, we found that newly emerged females showed no preference for either the control or the host arm (figure 2B). Five days later, they were strongly attracted to the host cues and this attraction was suppressed after their blood meal and until oviposition (magenta, figure 2B). This general trend in attraction to host cues tracks the trend in blood feeding. Interestingly, the response of virgin females to host cues in the Y-maze was highly variable under all conditions tested (cyan, figure 2B). Yet, like in the blood-feeding assay, bloodfed virgins were comparatively more attracted to human host cues than were the mated blood-fed females (figure 2B).

These data suggest that, like blood feeding, host seeking is also modulated through the female's reproductive cycle, and mating influences this modulation by enhancing it initially and suppressing it post-blood meal. The variability in response of virgin females suggests that mating enhances the female's initial attraction to a host.

Neurotranscriptomics reveals several candidate genes that promote blood feeding

While sensitivity to host kairomones may modulate blood feeding, it is also possible that central brain circuitry contributes to it. We reasoned that cues from mating, or signals from other tissues such as the midgut could act on brain circuitry to modulate blood feeding. To determine the molecular underpinnings of this behavioural modulation, we performed RNA sequencing from the central brains (brains without optic lobes) of mosquitoes at different stages of their reproductive cycle (consequently different states of blood-deprivation or -satiety). This included females at emergence (D0), 5-days after emergence (D5; sugar-fed, blood-deprived), virgins and mated females one day after a blood meal ($D1^{PBM}$, $D1^{PBM}$(m)), and females one day after oviposition ($D1^{PO}$(m)). Age-matched males were included for the earliest two time points (figure 3A).

For each of our samples, 81-87% of about 20-30 million reads aligned to the *An. stephensi* genome (66). The variance between our replicates was low as represented in the Principal Component Analysis (PCA) (figure 3B). In PCA, older female brains, regardless of mating-, blood feeding-, or ovipositionstatus clustered together while the youngest female and male brains stood apart on PC1 and PC2 suggesting that age and sex contribute most significantly to the variance in our data (figure 3B). The expression pattern of some hallmark genes showed the expected trends. Genes involved in eclosion behaviours – *eclosion hormone* (67) and *partner of bursicon* (68) – showed the highest expression in the D0 samples and were not expressed later. Male-specific genes, *HMG-B-13-like*, were expressed only in males (69), and genes involved in vitellogenic stages of egg-development – *vitellogenins, ecdysone receptor*, and *E-20-monooxygenase* (70)– were expressed only in post-blood meal samples (figure 3C).

We next explored this data for insights into the molecular underpinnings of blood-feeding modulation. Reports suggest that anticipatory metabolic changes in female germline cells and a 'vittelogenin wave' in fat bodies can drive specific appetite in *Drosophila* (71) and *Ae. albopictus* (48), respectively. However, we did not find any evidence in the neurotranscriptomes for a similar anticipatory metabolic change that might drive blood feeding in the mosquito brain (figure S2B). Instead, our data suggests altered carbohydrate metabolism after a blood meal, with the female brain potentially entering a state of metabolic 'sugar rest' while actively processing

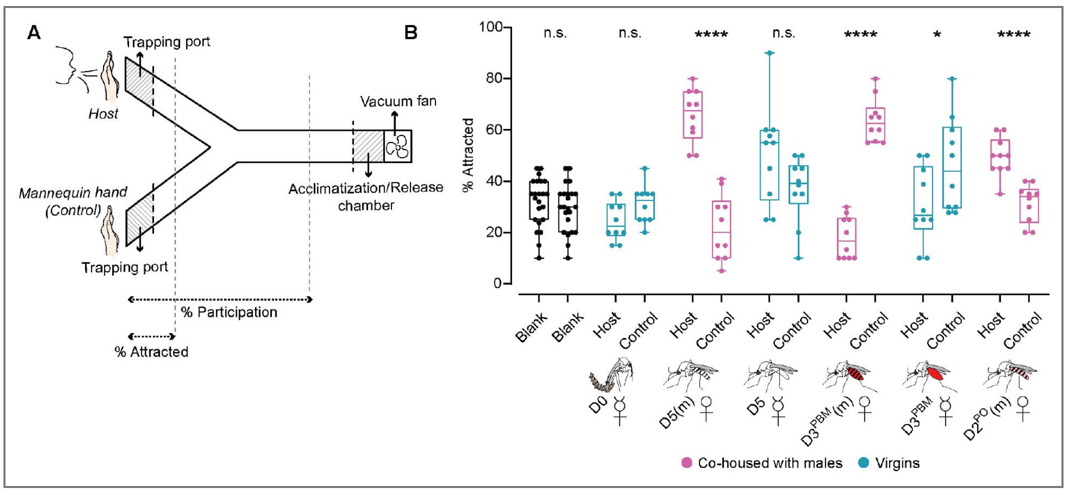


Fig. 2. **Host seeking is modulated like blood feeding is in *An. stephensi*.**

A. Y-maze schematic for host-seeking behaviour of female *An. stephensi*. Females were acclimatised in the chamber for 5 minutes while being exposed to host kairomones presented in a test arm. A fan sucked the air from both host and control arms at 0.3-0.6m/s. Post-acclimatisation, mosquitoes were released and allowed to choose between the two arms. **B**. Percent females attracted to either host cues or control arm at the indicated time points. 17-21 females/trial, n=10 trials/group. Unpaired t-test; n.s.: not significant, p>0.05.; * p<0.05; ****p<0.0001. (D0: day of emergence ; D5(m): 5 days post-emergence, mated; D5: 5 days post-emergence, virgin; $D3^{PBM}$(m): 3 days post-blood meal, mated; $D3^{PBM}$: 3 days post-blood meal, virgin; $D2^{PO}$(m): 2 days post-oviposition, mated)

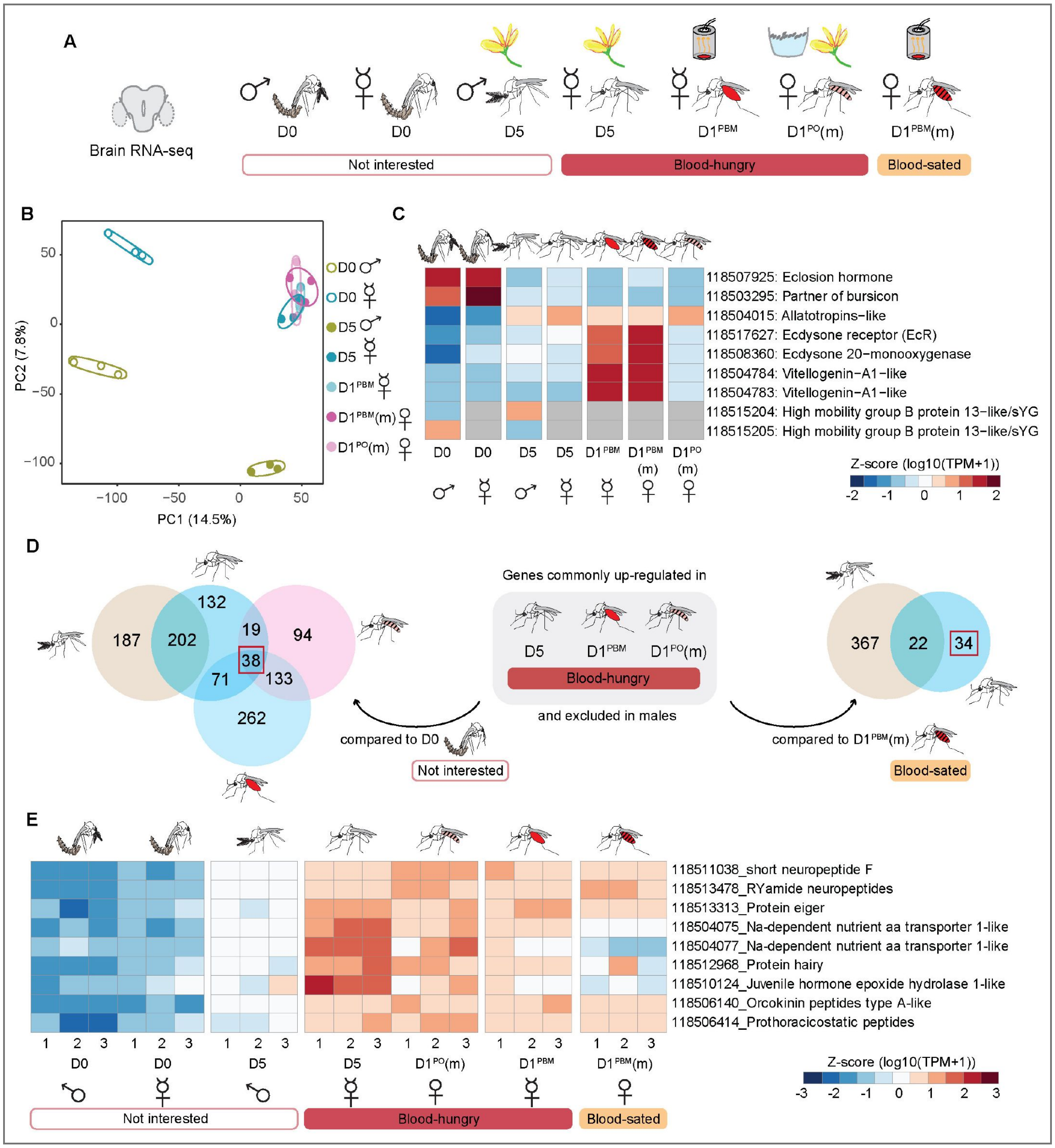


Fig. 3. Neurotranscriptome of *An. stephensi*.

A. Central brains sampled from males and females at different ages and feeding preferences for bulk RNA-seq. Uninterested in blood: D0 males, D0 females, D5 males. Blood-hungry: D5, sugar-fed virgin females, D1^{PBM} virgin females, and D1^{PO}(m) mated females. Blood-sated: D1^{PBM}(m) mated females. n=100 central brains per replicate; 3 biological replicates per sample, 7 samples. (m): mated. **B**. Principal Component Analysis (PCA) of the normalised counts-permillion. **C**. Genes (listed on the right) with known expression patterns plotted across different samples. Average of the triplicate data are represented as z-score normalised log10 (TPM+1) values. Grey cells represent no data. **D**. Identification of candidates potentially involved in promoting blood-feeding behaviour. Two types of comparisons were made: (left) genes commonly upregulated in the three blood-hungry conditions (see figure 3A) when compared against D0 females and excluded in males, and (right) genes upregulated in D5 sugar-fed virgin females when compared against D1^{PBM} mated females (blood-sated), and excluded in males. Number of candidate genes identified from two different sets of analyses are boxed in red. **E**. Nine candidate genes (listed on the right), shortlisted for further validation. RNA-seq data from central brain samples in triplicates are represented as z-score normalised log10 (TPM+1) values. (D0: day of emergence; D5: 5 days post-emergence, virgin; D1^{PBM}: 1 day post-blood meal, virgin; D1^{PBM}(m): 1 day post-blood meal, mated; D1^{PO}(m): 1 day post-oviposition, mated) proteins (Figure S2B, S3). However, physiological measurements of carbohydrate and protein metabolism will be required to confirm whether glucose is indeed neither spent nor stored during this period. We similarly plotted genes involved in lipid metabolism, neurotransmitters, neuropeptides and their receptors and found no significant patterns (figure S4, S5, S6). In summary, contrary to our expectations, we saw no evidence in the neurotranscriptomes for anticipatory metabolic changes in the brain that might in turn drive blood-feeding behaviour. It is possible, however, that other forms of regulation that do not involve transcriptional changes could still influence metabolism.

In an alternate approach to identifying genes that might modulate blood feeding, we took the intersection of genes that were enriched in the three blood-hungry conditions (and absent in males). These comparisons were made against the two conditions that were not motivated to blood-feed: females at emergence and mated blood-fed females (figure 3D). These shortlisted 38 and 34 genes respectively, gave us 68 unique candidates (excluding common genes and isoforms) that might promote blood feeding in *An. stephensi* (figure S7). Notably, we did not find any genes to be differentially expressed between the two blood-fed conditions (virgins and mated), despite their contrasting behaviours (figure S2A). The strongest signatures observed in these females were those related to metabolism (figure S2B). It is possible that the differences lie in small populations of cells, which get obscured in the averaged transcriptomes of the entire brain.

Neuropeptides *sNPF* and *RYamide* promote blood feeding in *An. Stephensi*

We employed multiple, independent criteria while further shortlisting candidates from the above analysis: 1) genes that showed the expected expression pattern: low in newly emerged males and females; high in blood-hungry females; low in blood-sated females, 2) all neuropeptides, since these are known to regulate feeding behaviours, and 3) all genes related to juvenile hormone (JH) and 20-hydroxyecdysone (20E) signalling pathway, since they regulate development and reproduction in mosquitoes. This gave us 9 candidates (figure 3E): Two sodium-dependent amino acid transporters *118504075 and 118504077*. Transporters can sense nutrients and regulate feeding in *Drosophila* (29, 72). The neuropeptides included *Orcokinin (Ork), Prothoracicostatic peptides (Prp), RYamide (RYa)* and *short neuropeptide F (sNPF)*, and a metabolic hormone *Eiger*, which responds to low nutrient conditions in *Drosophila* (73). *Protein Hairy* and *Juvenile hormone epoxide hydrolase 1-like (JHEH)* are involved in the JH pathway (74, 75)(figure 3E). Using dsRNA, we knocked each of these genes down and tested their blood-feeding behaviour. In each case, we ensured efficiency of gene knockdown post-hoc (figure 4A).

We first confirmed that the injection itself did not cause any behavioural changes (figure 4B) and in all following experiments either *dsGFP*-injected or uninjected females were included as controls.

We saw no change in behaviour upon loss of *Eiger*, *JHEH*, and *Ork* function (figure S8A-C, S8A'-C'). We were able to downregulate *Hairy* and *Prp* transcript levels by only 30-35%, hence cannot comment on their role in modulating blood feeding (figure S8D, S8E, S8D', S8E'). Downregulation of the two transporters was efficient and resulted in a marginal reduction in blood feeding (figure S8F, S8G, S8F', S8G'). However, their combined knockdowns, though effective, did not show any significant change in behaviour (figure S8H, S8H').

Two neuropeptides - *sNPF* and *RYa* - showed about 25% and 40% reduced mRNA levels in the heads but the proportion of females that took blood meals remained unchanged (figure S9B, S9C, S9E, S9F). We noticed, however, that many of these animals did not engorge and instead took smaller blood meals (figure S9D, S9G). Given this, and since neuropeptides are known to act in concert (76), we tested the possibility that *sNPF* and *RYa* might act together to regulate blood-feeding behaviour. Indeed, their simultaneous dsRNA injections resulted in about 40% of the animals not taking blood meals (figure 4C). Since RNA interference (RNAi) can be variable, and each female is an independent experiment, we pooled the females that took blood meals and those that didn't separately before quantifying mRNA from them. We noted that knock-down efficiency

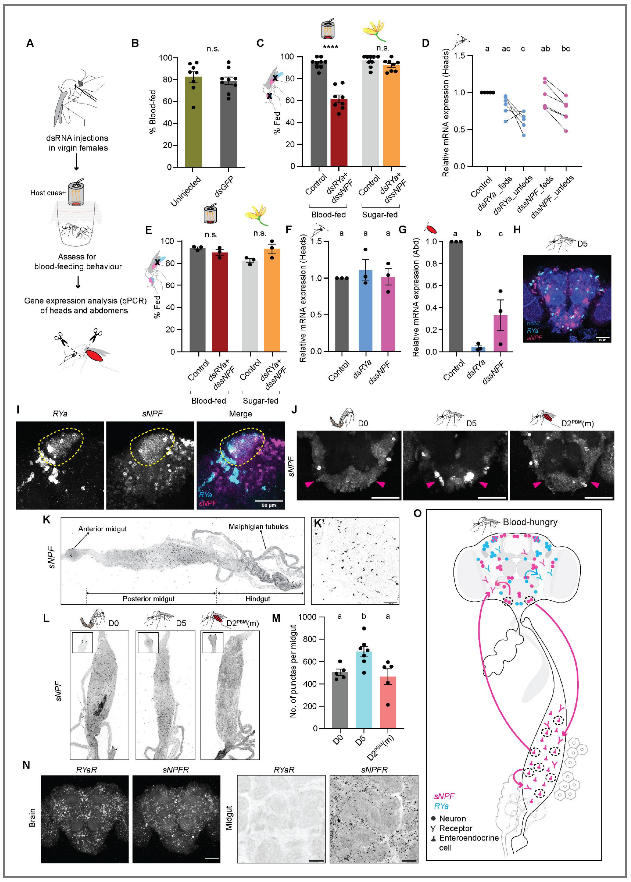


Fig. 4. ***short neuropeptide F (sNPF)*** **and** ***RYamide (RYa)*** **together promote blood feeding in** ***An. stephensi*****.**

A. Pipeline used for functional validation: dsRNAs were injected in adult virgins and tested for blood feeding. Knockdown efficiency was determined in heads and abdomens of fed and/or unfed females after the behaviour. **B**. Blood feeding in uninjected and *dsGFP*-injected females. 19-25 females/replicate, n=8-9 replicates/group. Unpaired t-test; n.s.: not significant, p>0.05. **C**. Blood- and sugar-feeding behaviour of females where both *RYa* and *sNPF* were knocked down in both the heads and abdomens. Controls: Uninjected females. 14-25 females/replicate, n=8-9 replicates/group. Unpaired t-test; n.s.: not significant, p>0.05; ****p<0.0001. **D**. Relative mRNA expressions of *RYa* and *sNPF* in the heads of *dsRYa+dssNPF* - injected blood-fed and unfed females assayed in (C). mRNA levels were compared to those in the uninjected controls. n=6 replicates/group. One-way ANOVA with Holm-Šídák multiple comparisons test, p<0.05. Data labelled with different letters are significantly different. **E**. Blood- and sugar-feeding behaviour of females when both *RYa* and *sNPF* were knocked down only in the abdomens. Controls: *dsGFP*-injected females. 17-21 females/replicate, n=3 replicates/group. Unpaired t-test; n.s.: not significant, p>0.05. **F**,**G**. Relative mRNA expressions of *RYa* and *sNPF* in the heads (F) and abdomens (G) of *dsRYa+dssNPF* - injected fed females assayed in (E). mRNA levels were compared to those in the *dsGFP*-injected controls. n=3 replicates/group. One-way ANOVA with Holm-Šídák multiple comparisons test, p<0.05. Data labelled with different letters are significantly different. **H**. *RYa* (cyan) and *sNPF* (magenta) HCR *in situ* hybridisation in central brain of 5 days-old sugar-fed virgin female. nc82 (blue). Scale bar, 50µm. **I**. Co-expression of *RYa* (cyan) and *sNPF* (magenta) in mushroom body Kenyon cells of 5 days-old sugar-fed virgin female. Scale bar, 50µm. **J**. *sNPF* expression in the SEZ of females uninterested in blood (D0), blood-hungry females (D5) and blood-sated females (D2^{PBM}(m)). Magenta arrow marks the novel *sNPF* cluster only in the blood-hungry (D5) condition. Scale bar, 50µm. **K**. *sNPF* HCR *in situ* hybridisation in gut of 5 days-old sugar-fed virgin female. Higher magnification image in shown in K'. Scale bar, 50µm. **L**. *sNPF* expression in the posterior midguts of females uninterested in blood (D0), blood-hungry females (D5) and blood-sated females (D2^{PBM}(m)). Note expression in enteroendocrine cells (K') and in two cells in the anterior midgut (L, insets). **M**. No. of *sNPF* positive cells. n=5-7 guts per condition. One-way ANOVA with Holm-Šídák multiple comparisons test, p<0.05. Data labelled with different letters are significantly different. **N**. *sNPF receptor (sNPFR)* and *RYa receptor(RYaR)* HCR *in situ* hybridisation of central brain (left) and midgut (right) of 5 days-old sugar-fed virgin female. Scale bar, 50µm. **O**. Proposed model: *RYa* and *RYaR* (cyan) are expressed only in the brain, while *sNPF* and *sNPFR* (magenta) are expressed both in the brain and the gut. Increased *sNPF* levels in both the tissues (dashed circles) promote a state of blood-hunger, which may drive feeding behaviour either by *sNPF's* action in the two tissues independently or via a communication between them. This happens in the context of *RYa* signalling in the brain. (D0: day of emergence; D5: 5 days post-emergence, virgin; D2^{PBM}(m): 2 days post-blood meal, mated) was higher in the heads of females that did not take blood meals suggesting that downregulating these genes caused the suppression of blood feeding (figure 4D). Importantly, since these females were offered both blood and sugar, we noted that their sugar feeding was unaffected (figure 4C).

Since RNAi results in systemic knockdowns and neuropeptides are known to influence behaviour from their action in non-neuronal tissues such as the gut (52, 76, 77), we modified our dsRNA injection protocol to knock *sNPF* and *RYa* down in the abdomen while leaving their levels in the head unaltered (figure 4F, 4G, S9J, S9L) (see methods for details). Abdomen-specific knockdown had no effect on blood feeding (figure 4E, S9I, S9K), indicating that abdominal *sNPF* and *RYa* alone cannot influence blood feeding and their action in the brain is necessary. In contrast, since every head knockdown of these neuropeptides also abolished their abdominal expression (figure S9C, S9F), we cannot rule out the possibility that both brain- and gut-derived neuropeptides modulate blood feeding. Indeed, since such neuropeptide-mediated gut-brain control of feeding is commonly seen (76), we propose a model, consistent with our data, where brain-derived *sNPF/RYamide* is essential to trigger blood feeding, while gut-derived peptide acts as a modulatory feedback signal. This remains to be tested with cell-specific tools that are currently not available for this species.

To determine in which cells *sNPF* and *RYa* act, we used hybridisation chain reaction (HCR) (78, 79) *in situ*. We found that their transcripts are expressed in several non-overlapping neuronal clusters distributed across many regions in the brain (figure 4H, S10A) and only in the Kenyon cells of the mushroom body were they co-expressed (figure 4I, S10B). This is reminiscent of their expression in *Drosophila* and *Bombyx* (80–82). Importantly, we found a cluster of cells in the SEZ that expresses the *sNPF* transcript only in the blood-hungry state (figure 4J, supplementary movie). Its transcripts were either absent or diminished from this cluster in newly emerged or sated females. Since the RNAi experiments suggested the abdominal involvement of these neuropeptides, we also looked for their expression in the midgut where we could detect only *sNPF* transcripts, and not those of *RYa* (figure 4K,K', S11A, S11B, S11E). Interestingly, *sNPF* transcript expression was also modulated in the midgut enteroendocrine cells (EECs) - hormone-producing cells of the gut. In these cells, *sNPF* transcripts were specifically increased in blood-hungry females when compared to newly emerged or sated females (figure 4L,M).

Neuropeptides work by acting on their receptors to trigger downstream signalling cascades in the cells that express the receptors. We identified putative receptors for *sNPF* and *RYa* (see methods) and found that both were expressed in several overlapping cell clusters in the brain (figure 4N, S10C, S10D). In the midgut, however, we could only detect *sNPFR* and not *RYaR* (figure 4N, S11C, S11D, S11F). Since these receptors were identified based on homology, their functional validation will be necessary to implicate them in blood-feeding behaviour.

Together, these data are consistent with a model where increased *sNPF* expression in both the brain and midgut might drive the blood-feeding behaviour, but that this occurs in the context of *RYa's* action in the brain (figure 4O).

Conclusions

Here, we report the blood- and sugar-feeding behaviours of *An. stephensi*. While sharing key features with other mosquito species, *An. stephensi* exhibits distinct regulatory patterns. Unlike *Ae. aegypti*, it does not require mating to initiate blood feeding, though the presence of males is necessary to sustain the enhanced state. In both species, blood feeding is suppressed after a blood meal and until oviposition, but this suppression is mating-dependent in *An. stephensi*. Since unmated *Ae. aegypti* females did not blood-feed in our assays, we cannot comment on whether this suppression also requires mating.

Mosquito feeding behaviour is highly context-dependent, shaped by physiological state, environment, and experimental design (36). Our comparisons between *Ae. aegypti* and *An. stephensi* were conducted under standardised conditions. While direct comparisons across studies must be made cautiously, patterns observed under particular conditions can offer meaningful insights. For example, virgins of other *Anopheles* species readily blood-feed (36, 83–85), suggesting that this is typical of the genus, whereas *Ae. aegypti* virgins do so only under specific conditions and these responses vary among strains (38) (and references within). Whether blood-fed *Aedes* virgins suppress subsequent feeding is likely also context-dependent with reports suggesting both that they do (86, 87) and do not (42). Post-blood meal suppression of host seeking - well characterised and robust in mated *Aedes* - appears weaker in mated *Anopheles* and can be lifted prior to oviposition (46). Indeed, we too observed a stronger post-blood meal suppression of feeding in *Ae. aegypti* than in *An. stephensi* (compare Fig. 1C to Fig. 1D). Few studies have investigated this suppression in anopheline virgins, but when they do, they report a weak suppression (88). Taken together, *An. stephensi* displays characteristics of both culicine and anopheline feeding patterns - its virgin feeding patterns align more closely with other anophelines, yet, like mated *Ae. aegypti*, it still exhibits clear suppression of blood feeding until oviposition.

What mechanisms might underlie such mating-induced modulation of blood feeding? In *Drosophila melanogaster*, mating induces a suite of behavioural changes - including increased protein, salt and sugar intake that are mediated by male-derived sex peptide that acts on central brain circuits in the female brain (4–8). These changes are anticipatory in nature rather than a homeostatic control of feeding behaviour to meet reproductive demands (7, 8). In mosquitoes, the molecular mechanisms driving post-mating behavioural changes are not likely to be the same (89). Instead, mating-derived modulators such as HP1 in *Ae. aegypti* (59, 90) and male-deposited steroid hormone (20-hydroxyecdysone) in *An. gambiae* (91, 92) have been shown to induce sexual refractoriness. Whether they also shape feeding decisions remains unknown.

We have identified two neuropeptides - *RYa* and *sNPF* - that act synergistically to promote blood feeding in *An. stephensi*. Our data are consistent with a model where an increase in *sNPF* levels in the brain - likely released by a cluster of neurons in the SEZ - as well as in the midgut, promotes a state of blood-hunger. Whether this reflects gut-to-brain communication or parallel tissue-local computations to modulate behaviours remains unclear, especially since we detected the transcript of both *sNPF* and its receptor in both tissues. Inter-organ signalling is well-established in feeding regulation (76, 93). For example, gut-derived NPF in flies can influence brain circuits to increase protein intake (77). More broadly, feeding behaviour is likely regulated by a distributed network of hormonal and neuronal cues across multiple organs. Signals from the ovary, fat body, and other nutrient-sensing tissues may also contribute, acting through both central and peripheral circuits. Many of these inputs remain to be identified.

It is possible that *sNPF* promotes blood feeding independent of *RYa*. However, we were unable to knockdown *sNPF* completely to test this. Its synergistic requirement with *RYa*, despite no apparent modulation of *RYa* or its receptor, could result from a general state of hunger that *RYa* promotes, in the context of which, we were able to uncover the blood-feeding role of the diminished *sNPF*. Since many neurons co-express these receptors, it is possible that neurons down-stream of *sNPF* or *RYa* expressing neurons integrate the two inputs to modulate behaviour.

We note that our findings differ from those reported in *Ae. aegypti*. In *Aedes*, *sNPF* and *RYamide* act as satiety brakes—peptide or receptor activation curbs host seeking (49, 50, 52) —whereas in *Anopheles*, our knockdowns show that the same neuropeptides act as a hunger signal, stimulating blood feeding. While these differences are interesting, in the context of their evolutionary histories, and the known differences in how neuropeptides can act differently across species, they are not entirely unexpected. *Aedes* and *Anopheles* lineages diverged ≈150-200 million years ago (94), and the two genera have since diverged in almost every facet of biology— diel activity, host preference, feeding and oviposition sites, egg desiccation tolerance, genome size, and chemosensory repertoires among other traits. Neuropeptide pathways are known to drive opposing behaviours in different species: *sNPF* is known to be orexic in *D. melanogaster, A. mellifera, L. decemlineata, B. mori*, and *P. americana*, while it has anorexic effects in *S. gregaria* (95). Against this backdrop, it is unsurprising that the same neuropeptide pathway can drive opposite behaviours in *Aedes* and *Anopheles*. These comparative studies underscore a key lesson for vector control strategies in general. A strategy that is effective in one mosquito species may inadvertently boost vectorial capacity in another. Behavioural interventions, chemical modulators, and even gene drives therefore require rigorous cross-species validation because results from one mosquito species cannot be assumed to hold true for another.

Methods

Mosquito culture

Anopheles stephensi (Indian strain; Bangalore strain, TIGS-1) and *Aedes aegypti* (Bangalore collection) were reared in BugDorm cages and maintained at 26-28°C, 70-80% relative humidity with a photo-period of 12h light: 12h dark. All the behavioural assays were performed at these conditions of temperature and humidity. Larvae were fed on a mixture of yeast and dog biscuit (1:3 ratio) and cultured in trays in appropriate densities to prevent over-crowding. Adult mosquitoes were provided *ad libitum* access to 10% sugar solution (8% sucrose +2% glucose +0.05% methyl paraben +5% vitamin syrup (Polybion SF Complete, Merck Ltd.)) at all times, unless specified otherwise. Females were provided O+ve human blood, secured periodically from blood banks, both for general maintenance and feeding assays.

Regulatory permissions

This project was approved by the Institutional Human Ethics Committee (Ref: inStem/IEC-22/02) and the Institutional Biosafety Committee (TIGS/M-7/8/2020-1).

Blood-feeding assay: first blood meal, 0-120h post-e-mergence (D0-D5)

All *An. stephensi* feeding experiments were performed from ZT12-ZT15 and *Ae. aegypti* experiments from ZT10-ZT12 (Zeitgeber time 0 is defined as the time of lights ON). 0-7h-old mosquitoes were cold anaesthetised and collected during the day in 1100ml paper cups covered with a net and secured with a rubber band. These were maintained at the conditions of temperature and humidity described above. To collect virgins, males were separated at the time of collection. 20 females with males (1:1 ratio) or without males were collected for each trial and were aged appropriately with *ad libitum* access to 10% sugar solution (supplementary figure S1A).

Females were assayed for their blood-feeding behaviour from the day of emergence (D0, 0-7h-old) to 5days of age (D5), every 24h. They were provided constant access to sugar, except when they were offered blood. A detailed schematic has been shown in figure S1A. On the day of the test, appropriately aged females were offered a blood meal via a Hemotek feeder (Hemotek Ltd), which was maintained at 37°C and perfumed with human skin odours. It has been reported that female mosquitoes need to be ‘activated’ to take blood meals. This involves the presentation of CO_2 (and other host cues) to the females, likely because CO_2 has been shown to gate temperature and visual cues (96, 97). So, prior to offering the blood meal, the mosquitoes were ‘activated’ by the presentation of a hand placed above the cup for 5 minutes, followed by 3 exhalations from the experimenter. Additionally, blood was supplemented with 1mM ATP (Sigma A6419) before loading the feeder (figure 1B). Females were allowed to feed for 60-90mins in dark and abdomens were scored visually for presence of blood, irrespective of the amount ingested (inset, figure S1A). For D0, no sugar was provided at the time of collection. Females were allowed to acclimatise for at least 30 mins in the cups and then offered a choice of both sugar and blood simultaneously, to assess their preference for blood soon after emergence (figure S1A).

In the case of co-housed females, males and females were housed together from the time of collection until the day of the experiment. Males were separated post-hoc and spermathecae of the females were dissected to score for the presence or absence of sperm (figure 1G). All the assays with virgin and co-housed females were performed in parallel by a single experimenter. Mosquitoes from at least 4 independent batches were tested for each day of the experimental timeline. For *Ae. aegypti*, both virgin and mated females (mating status was confirmed via post-hoc dissection of spermathecae and all co-housed females were found to be mated) were tested for first blood meal at only 8 days post-emergence (D8, figure 1C). Behavioural experiments were performed the same way as described for *An. stephensi*.

Blood-feeding assay: second blood meal, 24-96h post-first blood meal ($D1^{PBM}$-$D4^{PBM}$)

To ensure that the second blood meal is not driven due to an incomplete first blood meal, we selected only fully engorged females to test for subsequent feedings (figure S1A). 5 days-old *An. stephensi* (both virgins and co-hosued with males) and 7-9 days-old *Ae. aegypti* (mated, as previously determined) were fed on blood and fully fed females were separated and collected in paper cups as described above. As a female takes about 2 days to digest blood, the second blood meal ingested 24h post-first blood meal ($D1^{PBM}$) was detected by the presence of Rhodamine B (Sigma R6626), which fluoresces in the red range (figure S1A). Spiking the second blood meal with Rhodamine B (0.04μg/μL of blood) allowed visualisation of the freshly ingested blood, without affecting the feeding efficiency (supplementary figure S1B-D). To assess feeding at subsequent days ($D2^{PBM}$-$D4^{PBM}$), abdomens were dissected to differentiate between the bright red freshly ingested blood from the dark and clotted first blood meal. In the case of co-housed *An. stephensi* females, spermathecae were not dissected to confirm mating as females were presumed to be mated when collected for this assay (5 days-old and co-housed with males for the entire duration from the time of emergence), as previously determined in figure 1G. To confirm whether mating suppresses subsequent feeding (figure 1H), fully fed virgins were collected and allowed to mate for 3-4 days with aged-match males. No males were introduced in the virgin-only controls. Blood-feeding assay was performed as described above. Spermathecae were dissected post-hoc to determine the mating status and unmated females were not considered for the final analysis. Datasets with less than 80% total mating were discarded.

Blood-feeding assay: post-oviposition, 24-48h post-egg laying ($D1^{PO}$-$D2^{PO}$)

For blood-feeding behaviour post-oviposition, fully fed females were separated and a water cup lined with moist paper (ovipositor) was provided for egg laying 2 days post-blood meal (determined in a separate assay described below, supplementary figure S1E; figure S1A). Females were allowed to lay eggs overnight. The following morning, non-gravid females were visually identified by their lean bellies and separated in cups. Blood-feeding assay was performed the same evening for 1 day post-oviposition ($D1^{PO}$) and the following day ($D2^{PO}$), as described above. Females were dissected post-hoc and scored for absence of mature ovaries, as a sign of successful oviposition. Gravid females were not considered for the analysis.

Determination of oviposition timing for *An. Stephensi*

To determine the day of maximum oviposition, 5-6 days-old females co-housed with males were fed on blood, as described above. Fully fed females were separated and collected in BugDorm cages in groups of 10 females/cage. An ovipositor was provided the same day and females were allowed to lay eggs overnight. Eggs laid were counted the next day and a fresh ovipositor was provided. This cycle was repeated for the next 5 days. Total eggs laid each day were divided by the number of females present at the time of counting (supplementary figure S1E).

Sugar-feeding assay for *An. Stephensi*

As mosquitoes feed on sugar in small bouts, as opposed to a replete blood meal, we assessed sugar-feeding behaviour of females in windows of 24h for each test day. 0-7h-old mosquitoes were collected during the day, separated in cups and aged appropriately. They were given access to 10% sugar solution at all times, as described above, except for the day of emergence (D0) when no sugar was provided at the time of collection (similar to the blood-feeding assay performed at D0).

To assess their sugar-feeding behaviour on the day of emergence (D0), 0-7h-old females were acclimatised in the cups for 1h and sugar spiked with Rhodamine B (1mg/25ml of sugar; Sigma R6626) was presented for 2.5-3h. The fluorescence of Rhodamine B in the red channel allowed visualisation of even the tiniest amounts of sugar ingested by a female (figure S1G). For D1 feeding, 3-10h-old females (with access to normal sugar so far) were then provided with Rhodamine B-sugar and allowed to feed on it for 24h. For D2 feeding, 27-34h-old females (with access to normal sugar so far) were fed on Rhodamine B-sugar for the next 24h and scored as D2. This cycle was repeated until D5. In the case of co-housed females, males were separated post-hoc and spermathecae of the females were dissected to score for the presence or absence of sperm (figure S1H).

To assess sugar feeding post-blood meal, females fully fed on blood were collected and separated in cups, as described above. They were allowed to feed on Rhodamine B-sugar for 24h following a blood meal and scored as $D1^{PBM}$. The next set of females (with access to normal sugar so far) were then provided with Rhodamine B-sugar from 24h-48h post-blood meal and scored as $D2^{PBM}$. This cycle was repeated until 4 days post-blood meal ($D4^{PBM}$, figure 1E).

For sugar feeding post-oviposition, blood-fed females were provided an ovipositor 3 days later and were forced to oviposit during daytime by artificially creating dark conditions. Oviposited females (non-gravid) were then visually identified, separated and presented with Rhodamine B-sugar for the following 24h and scored as $D1^{PO}$. The next set (with access to normal sugar so far) was then provided with Rhodamine B-sugar, 24-48h post-oviposition and scored as $D2^{PO}$(figure 1E). For each day tested, females were dissected post-hoc and scored for absence of mature ovaries, as a sign of successful oviposition. Gravid females were not considered for the analysis.

Dual choice assay of blood and sugar

Mosquitoes were collected as described above. Males and females were co-housed for 4-6 days with *ad libitum* access to sugar. To assess the choice between sugar and blood, mosquitoes were either given continuous access to normal sugar (sugar-sated) or starved with water 24h prior to the assay (sugar-starved). At the time of the assay, post-activation (as described above), blood and Rhodamine B-spiked sugar (1mg/25ml of sugar, Sigma R6626) were provided in parallel. After 60mins, abdomens were scored visually for ingestion of blood under white light, and under fluorescent microscope (red channel) for ingestion of sugar (figure S1F). Co-housed males served as an internal control for sugar feeding. As males are unable to feed on blood, they were only scored for presence or absence of Rhodamine B-spiked sugar. Results are presented in figure 1F.

Statistical analysis of feeding behaviours

Statistical analysis was performed using R (version 4.4.2) and Graph-Pad Prism (v10.2.0). Blood- and sugar-feeding behaviours of virgins and co-housed females in figure 1D, E, were analysed using Generalized Linear Mixed-effects Models (GLMMs) with a binomial distribution and logit link function. The models were fitted using the “glmer” function from the “lme4” package in R. The probabilities of blood- or sugar-feeding, both pre- and post-blood meal were modeled as a function of Day (age) and Group (virgin vs. co-housed with males), including their interaction term to test whether group differences varied across days. Replicate identities (BioID) and larval batches (only for first blood meal in sugar-fed females) were included as random factors to account for replicate-level and batch-level variations respectively. Post-hoc analyses were conducted using estimated marginal means (emmeans package) for age-effects within groups (p-value adjusted for multiple comparisons using Tukey’s method) and pairwise comparison between groups at each day (p-value adjusted using Bonferroni method). To test whether mating success altered the sugar- and blood-feeding propensity of females co-housed with males, the probabilities of feeding were modelled only the co-housed data, as a function of Day (age) and Mating (yes or no), including their interaction term and accounting for variation due to replicate IDs (BioID). Statistical significance was set at $\alpha = 0.05$ for all analyses. Complete model outputs are provided in supplementary sheet 1.

Statistical analyses for data in Figures 1C, 1F, and 1H were performed using GraphPad Prism (v10.2.0) with appropriate tests as indicated in figure legends.

Y-maze olfactory assay for host-seeking behaviour in *An. Stephensi*

A 30inch Y-maze was constructed as per the WHO guidelines (*https://www.who.int/publications/i/item/9789241505024*). Each arm of the Y-maze had a trapping port, while the stem was attached to a holding port. Each port had a mesh screen at one end, while the other was fitted with a sliding door to trap or release mosquitoes. The holding port was attached to a fan which pulled air from the two arms, over the mosquitoes, and out at a velocity of 0.3-0.6m/s. Optic flow information (alternate black and white stripes) was placed underneath the Y-maze to act as visual guide for navigation.

Following conditions were tested: D0 virgins, D5 sugar-fed virgins, D5 sugar-fed mated females (D5(m)), 3 days post-blood meal virgins ($D3^{PBM}$), 3 days post-blood meal mated females ($D3^{PBM}$(m)), and 2 days post-oviposition mated females ($D2^{PO}$(m)). For mated conditions, females were co-housed with males for 5 days post-emergence and assumed to be mated based on our previous findings (figure 1G). For each condition, mosquitoes were collected and provided *ad libitum* access to sugar as described above. Females were transferred to holding ports 7-8h prior to the assay and were starved during this time with only access to water. For each run, mosquitoes were acclimatised for 5 minutes: 2 min with clean air and 3 minutes with an experimenter presenting a hand at one of the arms and simultaneously exhaling at resting rates. To account for disruption in airflow, the hand of a mannequin was placed at the other arm (control). Postacclimatisation, the door of the holding port was opened to release the mosquitoes and they were allowed to make a choice for 5 minutes in the presence of continued host cues. The ports were then closed and mosquitoes in each arm counted. All the mosquitoes trapped in a port, sitting on the door or resting on the half-length of the Y-arm were considered as "attracted" to that stimuli (host or control, figure 2A). Participation percentage was also calculated as number of mosquitoes that passed half-way through the Y-stem/total number of mosquitoes released (figure S1I).

Several controls were included to avoid experimental artefacts. Each experimental day started with a blank trial (where no stimuli were presented on either arm of the Y-maze) and the side of the host cues presentation was alternated with each trial to eliminate side biases. Trials for blood-fed virgins and mated females were done both in parallel and on multiple days to eliminate the influence of external factors which might be present on a given day. When testing blood-fed females, age-matched sugar-fed females (blood-hungry) were included wherever possible as positive controls. These females consistently showed attraction to host cues, as expected. Lastly, we made sure that at least 50% of the mosquitoes had made either choice in all runs, except for D0 (newly emerged females) and blank runs. In the rare cases that this was not true, we did not consider those for the analysis. Statistical analysis was performed using GraphPad Prism (v10.2.0). To compare the attraction of females to host cues and control in figure 2B, each time-point was treated as an independent experiment and p-values were computed for each pair-wise comparison.

Bulk RNA sequencing of central brains and data analysis

For bulk RNA sequencing, central brains (ie. brains without optic lobes) were dissected from the following conditions: D0 and D5 sugar-fed males; D0 and D5 sugar-fed virgin females; 1 day post-blood meal virgin ($D1^{PBM}$) and mated females ($D1^{PBM}$(m)) and 1 day post-oviposition mated females ($D1^{PO}$(m)). For post-blood meal conditions, 5 days-old females were fed on blood and fully fed females were separated for sample preparation. Females co-housed with males for 5 days were assumed to be mated based on our previous findings (figure 1G). Three replicates from independent batches were prepared for each condition.

Mosquitoes were cold anaesthetised before dissections. Central brains (with optic lobes removed) were dissected in RNase-free 1X PBS and immediately placed into ACME solution (13:3:2:2 of water, methanol, acetic acid, and glycerol) on ice, which mildly fixes the tissues (98). For each condition, 100 brains were dissected per replicate. The mildly fixed brains were washed with cold 1X PBS and centrifuged at 13,000 rpm for 5 min at 4°C to remove ACME. Brains were then homogenised in TRIzol (Invitrogen), flash-frozen, and stored at -80°C until RNA extraction. RNA was extracted using Direct-zol RNA MicroPrep (Zymo Research) columns. RNA quantity and quality were assessed using the ThermoFisher Qubit 4 and RNA 6000 Nano kit Bioanalyser, respectively.

The cDNA library was prepared for each sample using the NEBNext Ultra II Directional RNA Library Prep kit (NEB E7765L), with an average fragment size of 250 bp. Some of the library samples were subjected to size and quality checks using a bioanalyser. Paired-end sequencing was carried out on the Illumina Hiseq 2500 sequencing platform, generating 2×100 bp reads. The library preparation and sequencing were performed at the Next Generation Genomics Facility (NGGF) at Bangalore LifeScience Cluster (BLiSC), India. Post-sequencing, samples were demultiplexed and delivered as raw Fastq files. After passing the quality check using the FastQC tool, the reads were mapped onto the reference *An. stephensi* transcriptome (NCBI RefSeq assembly GCF013141755.1) (66) using Kallisto v0.43.1 (99) with default parameters. 81-87% of reads aligned to the mosquito transcriptome.

Transcript-level abundance was quantified using Kallisto and gene-level summarisation was performed using the Tximport function in R, with transcript isoform information retained (supplementary sheet 2). Using the EgdeR package, the raw counts were converted to counts per million (CPM) and log2 transformed before data filtering (genes with less than 1 CPM in at least 3 or more samples were filtered out) and normalisation, using the trimmed mean of M-values (TMM) method. Principal component analysis (PCA) was performed using the prcomp function in R and visualised using the ggplot function. Data was checked for batch effects using gPCA (100) in R and the test statistic “delta” was found to be 6.957686e-05, with a p-value of <0.001, indicating no batch-effects.

Differential gene expression analysis

Differential expression analysis was performed using the Limma package in R. A design matrix was created to specify the sample conditions, and the Voom function was used to estimate the mean-variance relationship across genes. A linear model was then fitted to the data using the mean-variance relationship. Bayesian statistics were calculated for the data by creating a contrast matrix to extract the linear model fit. To identify differentially expressed genes (DEGs), the “decide-Tests” function was used by applying a fold change threshold of 2 (log2fc=1) and a FDR (false discovery rate) adjusted p-value cut-off of 0.05. The number of genes identified to be upregulated and downregulated in each pair-wise comparison are shown in figure S2A and the gene lists are provided in supplementary sheet 3.

To plot transcripts per million (TPM) values as heatmaps, the Fastq files for the central brain samples were mapped onto the reference *An. stephensi* transcriptome and abundance was quantified using Kallisto the same way as described above. Gene-level summarisation was done using Tx-import as described above, except all the isoforms of a given transcript were mapped to a single gene (supplementary sheet 2). The resultant TPM values were first filtered for TPM>0.1 in each sample. Next, the genes that were not quantified in at least 2 of the 3 replicates of each sample group were removed. The filtered dataset were then log10 transformed and heatmaps were plotted using the pheatmap function in R. To construct heatmaps of genes involved in carbohydrate and lipid metabolism (supplementary figure S3, S4), key enzymes of the respective canonical pathways were manually curated based on the existing literature. Gene annotations for neuropeptides and neuropeptide receptors were curated from published sources (101–103). Fat body, midgut and ovary data (SRX620223, SRX618937 and SRX620224; bioproject SRP043489) for 2-4 days-old, sugar-fed *An. stephensi* females were taken from Prasad et al.(2017) (104). Fastq files were downloaded from the SRA explorer and processed similarly (supplementary sheet 2).

Gene set enrichment analysis

eggNOG mapper tool 2.1.9 (105) (parameters: 70% identity and minimum 70% query coverage) was used to find functionally annotated orthologs of the identified DEGs in *An. gambiae. An. gambiae*-specific orthologs were fed into the GOSt tool from gProfiler2 (106) toolset for functional enrichment analysis. A custom gene list comprising of all the genes identified in our RNA-seq experiment was provided as a background instead of the whole genome. Using this background, an ordered query was run with an FDR adjusted p-value threshold of 0.05 to determine which GO terms (Biological Processes) were over-represented. This was done for all sets of DEGs and the top 10 terms of each set were plotted using ggplot function in R (figure S2B).

dsRNA-mediated gene knockdown

To synthesise the double-stranded RNA (dsRNA), coding regions (500-600bp) of candidate genes were amplified from cDNA of *An. stephensi* females using forward and reverse primers with T7 promoter sequence at 5' ends. The gel purified fragments were then used as a template for *in vitro* transcription reactions using Megascript RNAi kit (Invitrogen) or High-Yield T7 RNAi Kit (Jena Biosciences) as per the manufacturers' protocols. The dsRNAs were purified using the phenol-chloroform method and pellet dissolved in nuclease-free water. The dsRNA for green fluorescent protein (GFP) was synthesised from plasmid pAC5.1B-EGFP (Addgene 21181) to be used as a control for injection-related behavioural changes.

Virgins were collected for injections as described above. The day and amount of dsRNA to be injected were determined for each target gene individually. *dsEiger*: 2.5-3.6μg injected per female in 24h-old virgins; *dsHairy*: 2-3μg injected per female in 24h-old virgins; *dsJHEH*: 2.5-3μg injected per female in 72h-old virgins; *dsPrp*: 1.5-3μg injected per female in 72h-old virgins; *dsOrk*: 2.5-3μg injected per female in 96h-old virgins; *ds118504077* and *ds118504075*: 3-3.6μg injected per female in 96h-old virgins. For dual injections of *ds118504077+ ds118504075*, 3.5-4.2μg of a 1:1 mixture of purified dsRNAs was injected per female in 96h-old virgins.

To achieve tissue-specific knockdown (abdomen only or head+abdomen) of *sNPF* and *RYa*, injections were performed at different concentrations and on different days. Injecting dsRNA into 0-10h-old females produced abdomen-specific knockdowns without affecting head expression, whereas injections into 96h-old females resulted in knockdowns in both tissues. Moreover, head knockdowns in older females (96h-old) required higher dsRNA concentrations (3.5-4.2μg injected per female), with knockdown efficiency cor-relating with the amount injected. In contrast, abdominal knockdowns in younger females (0-10h-old) could be achieved even with lower dsRNA amounts (2-2.4μg or 3.5-4.2μg injected per female). For dual injections, 3.5-4.2μg of the 1:1 mixture of purified dsRNAs was injected per female as described above: in 0-10h-old females for abdomen-specific knockdowns and in 96h-old virgins for head+abdomen knockdowns.

Injections were performed on a FemtoJet 4i system (Eppendorf) and each female was injected with 0.5-0.6μL of the respective concentrated dsRNAs in the upper thorax. In some cases, dsRNAs were also injected in the abdomens to ensure that the site of injection does not alter the behaviour. Uninjected or *dsGFP*-injected (same concentration as that of the candidate gene) females were included as a control each time. 15-25 females were injected for each trial and at least 2 trials from independent batches of mosquitoes were performed for each target gene.

Injected females were allowed to recover in paper cups with *ad libitum* access to sugar and assessed for blood-feeding behaviour at 3 days post-injection, as previously described. Fed and unfed females were visually scored and 5-6 females were randomly selected for each sample-whole heads and abdomens were collected separately for analysis. Total RNA was extracted using Direct-zol RNA MicroPrep (Zymo Research) columns, as per the manufacturer's protocol. RNA was quantified using Qubit Broad range assay (Invitrogen). For each sample, 200-220ng RNA was used to prepare cDNA using the SuperScript™ IV Reverse Transcriptase (Invitrogen 18090050) or Maxima H Minus Reverse transcriptase (Thermo Scientific EP0752). Quantitative PCR (qPCR) was performed using PowerUp™ SYBR™ Green Master Mix (Applied Biosystems) to determine the knockdown efficiency. Relative mRNA abundances were calculated using the delta-delta Ct method (107), normalised to RpS7. Statistical analyses were performed using GraphPad Prism (v10.2.0). All the primers used in the study are listed in supplementary table 1.

To assess the sugar-feeding behaviour of dsRNA injected females (*dssNPF, dsRYa* and *dsRYa+sNPF*), injections were performed as described above. Injected females were allowed to recover with *ad libitum* access to normal sugar for the first two days, which was then replaced with Rhodamine B-spiked sugar (1mg/25ml of sugar, Sigma R6626) for the next 24h. Control females were treated similarly. The blood-feeding assay was performed 3 days post-injection as described above, with both blood and sugar provided in parallel. Abdomens were scored visually for ingestion of blood under white light, and under fluorescent microscope (red channel) for ingestion of sugar (figure S9A).

Hybridisation chain reaction (HCR)

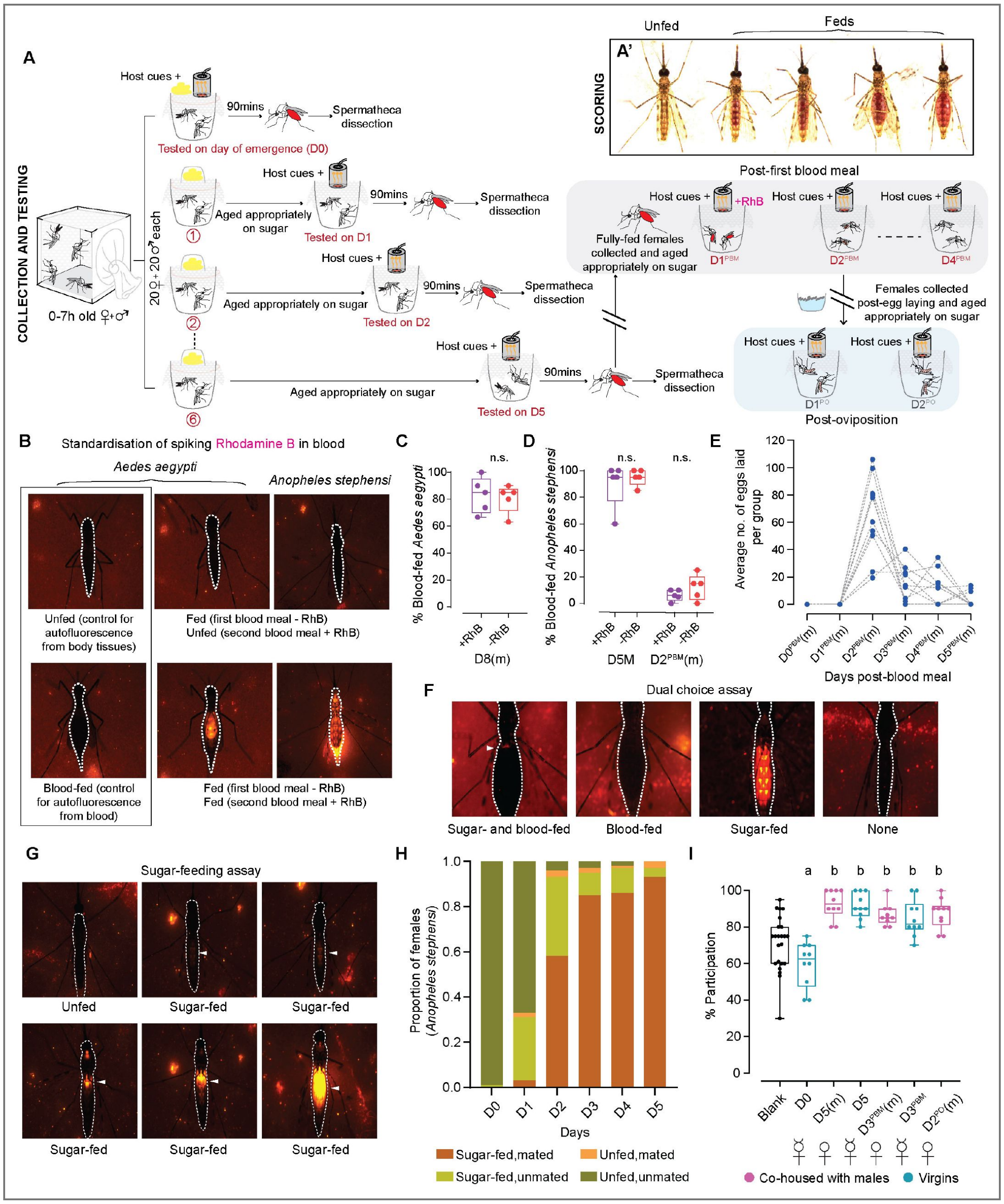
Previously published protocol (108) was modified to perform HCR *in situ* hybridisations on brain and gut tissues collected from females under three different conditions: D0 (not interested in blood), D5 virgins (blood-hungry) and mated D2^{PBM}(m) (blood-sated). All the reagents, including the probes, amplifiers and buffers were procured from Molecular Instruments Inc. A set of custom-designed 20 probe pairs were used per target gene: *sNPF (118511038), RYa (118513478)* and their receptors. To find *sNPF* and *RYa* receptors in *An. stephensi*, putative protein sequences were blasted against orthologs in *Drosophila melanogaster, Ae. aegypti* and *An. gambiae*. Based on high identities, consistent in all three organisms, *118517381* and *118507167 (NPY receptor type 2)* were identified as *RYa receptor (RYaR)* and *sNPF receptor (sNPFR)*, respectively.

5-7 brains for each target, per condition were dissected in ice-cold PBS and fixed with 4% PFA (in PBS) for 20 minutes at room temperature. Post-washing steps with PTw (0.1% Tween-20 in PBS), 4 times 10 mins each, tissues were permeabilised using detergent solution (50mM Tris-Cl, pH7.5, 150mM NaCl, 1mM EDTA, pH8, 0.5% Tween-20, 1% SDS) for 30 mins at room temperature. Brains were then prehybridised in probe hybridisation buffer for 30 mins at 37°C, followed by incubation with neuropeptide probes (0.8pmol each) at 37°C for 48h in a humid chamber. Samples were washed with pre-warmed wash buffer, 4 times 15 mins each, at 37°C to remove unbound probes. This was followed by 2 washes, 5 mins each with 5X SSCT (0.1% Tween-20 in 5X SSC) at room temperature. Samples were then pre-amplified using amplification buffer for 30 mins at room temperature. For each sample, 2μL (3μM stock) of each hairpins were mixed and prepared by heating at 95°C for 90secs for annealing. They were allowed to cool to room temperature for 30mins in a dark chamber and 100μL amplification buffer was added to the mix. Brains were incubated in the hairpin solution for 12-16h at 37°C in dark and excess hairpins were washed off with multiple 5X SSCT washes: twice for 5 mins, twice for 30mins, once for 5 mins. For nc82 staining, all the steps were performed in 0.5% PTw buffer (0.5% Tween-20 in PBS). Brains were equilibrated in 0.5% PTw and incubated with anti-mouse nc82 (1:200; DSHB, *RRID:AB2314866*) for three overnights at 4°C. Post-washing, goat anti-mouse Alexa-fluor405 (1:400, Invitrogen) was added and samples were left in dark, for one-two overnights at 4°C. After washing off the unbound secondary antibody, samples were equilibrated for at least an hour in Vectashield and mounted in Vectashield for imaging. *In situs* against neuropeptide receptors were performed the same way, except the washing steps to remove the unbound probes were performed at 42°C and samples were incubated with the hairpin solution at room temperature (to minimise background). As controls for background signal, samples were prepared the same way, except no probes were added in the hybridisation buffer (figure S10E). Samples for all three conditions were dissected, processed and imaged the same day.

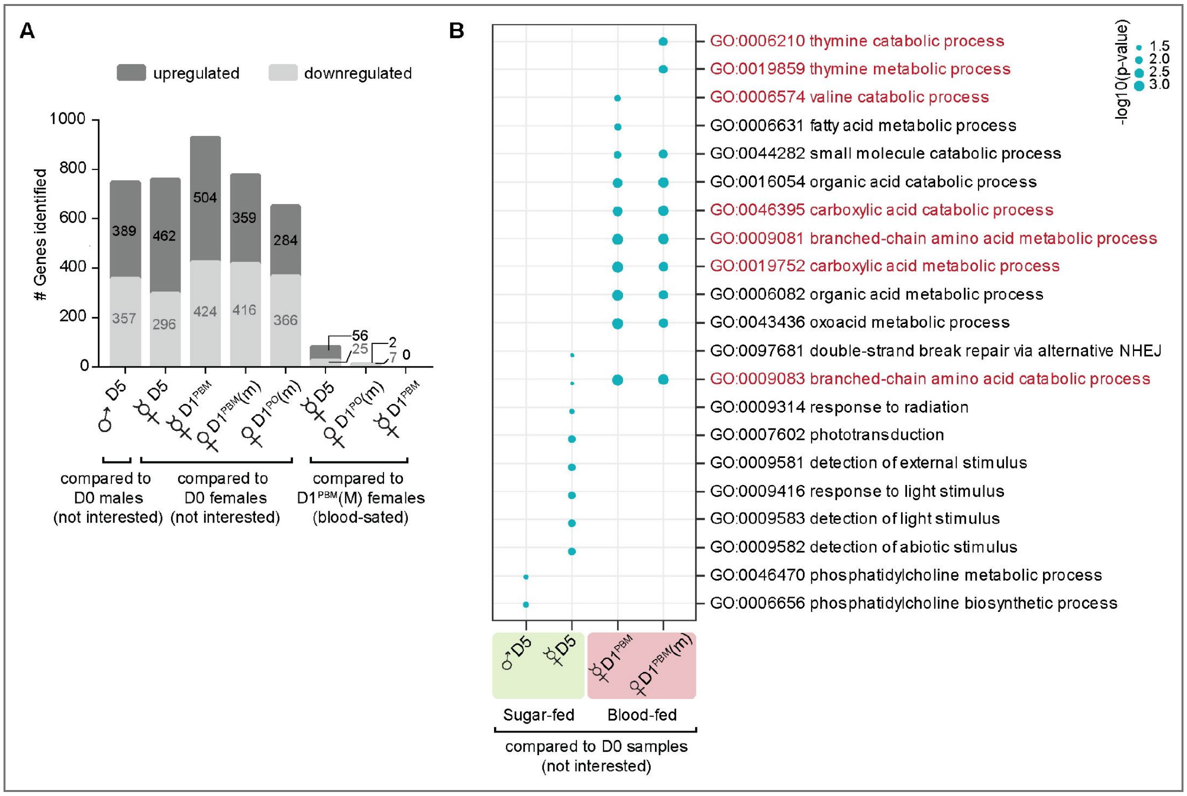
Protocol for gut *in situs* was adapted from Slaidina et al. (109). 5-8 gut samples for each target, per condition were dissected in ice-cold PBS and fixed with 4% PFA (in PTw) for 20 minutes at room temperature. Post-washing steps with PTw, tissues were dehydrated on ice with graded methanol washes: 25% methanol (in PTw), 50% methanol (in PTw), 75% methanol (in PTw) and 100% methanol for 10 mins each. Samples were stored in 100% methanol for up to a month, at -20°C. For rehydration, samples were washed sequentially with graded methanol: 75% methanol (in PTw), 50% methanol (in PTw), 25% methanol (in PTw), followed by a wash in permeabilisation solution (1% Triton-X in PBS). Samples were permeabilised for 2h at room temperature and fixed again with 4% PFA (in PTw) for 20 minutes. Following 10mins washes: washed once with PTw, twice with 50% PTw/50% 5X SSCT and twice with 5X SSCT, samples were pre-hybridised in probe hybridisation buffer for 30 mins at 37°C, followed by incubation with probes (0.8pmol each) at 37°C for 48h in a humid chamber. Post washing steps with wash buffer at 37°C and 5X SSCT at room temperature, samples were pre-amplified in amplification buffer for 30 mins at room temperature. Hairpins were prepared the same way as described above and samples were incubated in hairpin solutions for 12-16h in dark at room temperature. Following 2 washes 5 mins each with 5X SSCT, samples were incubated with DAPI (1:1000 in 5X SSCT) for one hour at room temperature. Samples were washed with 5X SSCT, 4 times, 10mins each and equilibrated in Vectashield for at least an hour before mounting. As controls for background signal, samples were prepared the same way, except no probes were added in the hybridisation buffer (figure S11G, H). Samples for different conditions, prepared on different days and stored at -20°C, were processed and imaged together. Statistical analyses were performed using GraphPad Prism (v10.2.0).

All samples were imaged at Central Imaging and Flow Cytometry Facility (CIFF) at the Bangalore Life Science Cluster (BLiSC), India. Images were acquired with a 40X/1.3NA or 10X/0.4NA (whole guts) objectives at a resolution of 512X512 pixels, on an Olympus FV3000 system. Images were viewed and manually analysed using ImageJ/FIJI (110). Maximum intensity projections of all z-stacks in an image are shown, except in figures 4H, 4I, S10B, S10D, where stacks were manually selected to capture relevant brain structures.

Supplementary figures and table

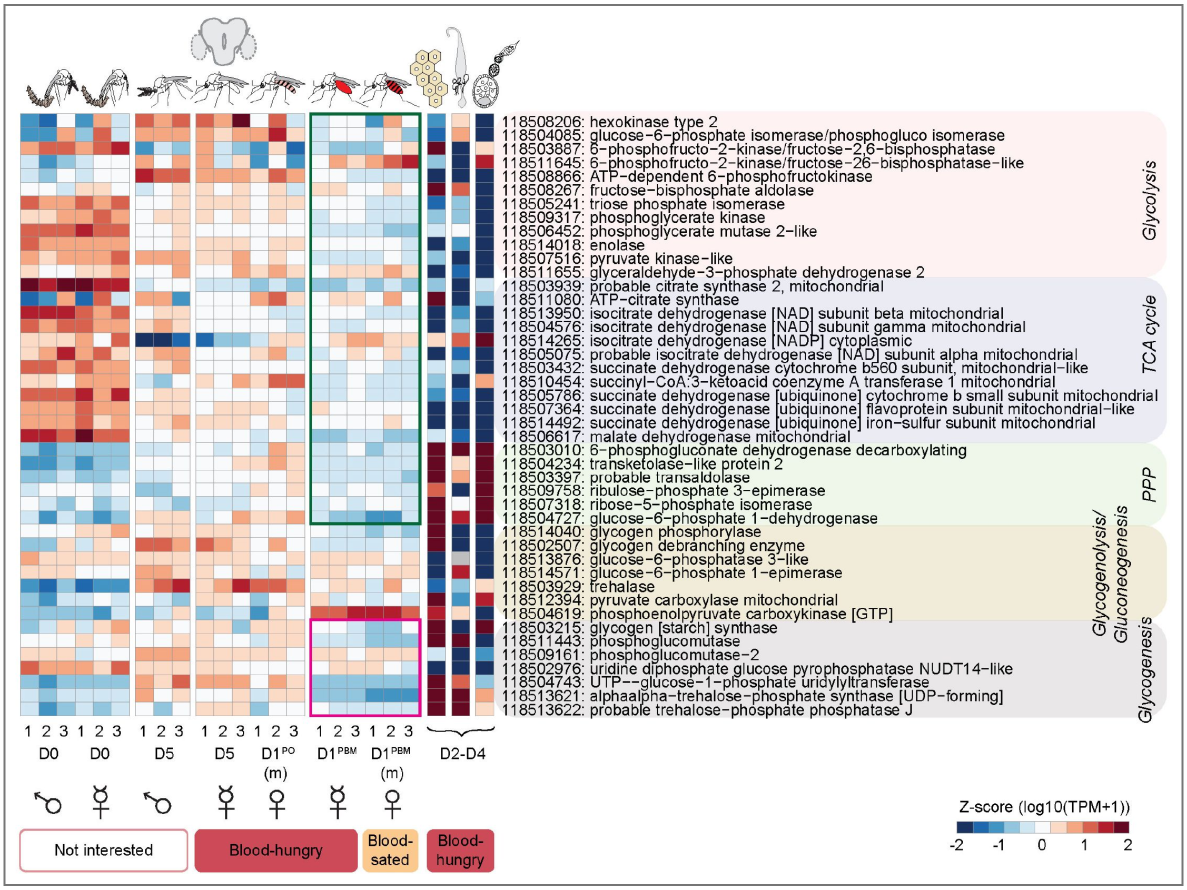


Supplementary Figure 1. Standardization of various behavioural assays to study feeding behaviours in *An. stephensi*. A. Detailed schematic of the assay designed to assess the blood-feeding behaviour of virgins and co-housed females across the reproductive cycle. 0-7h-old mosquitoes were collected and about 20 females (with or without 20 males) were separated into cups. Each cup was aged appropriately and tested on the relevant day (see methods for more details). Number of blood-fed mosquitoes were visually scored, irrespective of the quantity fed. Representative images of differently fed females are shown in A'. Females were tested for first blood meal from the day of emergence (D0) to five days of age, every 24h. Spermathecae of co-housed females were dissected post-hoc to determine the mating status. To assay for the second blood meal, fully fed females were collected and aged appropriately in groups of 20. To assess feeding 1 day post-first blood meal (D1^{PBM}), Rhodamine B was spiked in the second blood meal offered to the females. The subsequent days (D2^{PBM}-D4^{PBM}) were tested without any such spiking (grey box; see methods for more details). To assay for blood-feeding behaviour in females after oviposition (D1^{PO}-D2^{PO}), freshly oviposited females were collected and aged in groups of 20 for the appropriate time in cups (blue box; see methods for more details). In all cases, females were presented with host cues prior to the assay and testing was performed as shown in figure 1B (see methods for more details). **B**. Standardisation of spiking Rhodamine B dye in blood to assess feeding 24h post-first blood meal ($D1^{PBM}$). To assay for the second blood meal, blood-fed females were collected, aged for 24h and offered Rhodamine B-spiked blood in a Hemotek. All other assay parameters remained the same as for the first blood meal (see methods for more details). Widefield fluorescent images of *Ae. aegypti* and *An. stephensi* fed on second blood meal spiked with Rhodamine B are shown. Images on the left (black box) show mosquitoes not fed and fed on un-spiked blood (without Rhodamine B), as controls for autofluorescence from body tissues and blood itself, respectively. Fluorescence from Rhodamine B added to the second blood meal enabled identification of fresh blood intake in previously fed mosquitoes reliably. **C**,**D**. Comparison of feeding behaviours of *Ae. aegypti* females (C) and *An. stephensi* females (D) fed on blood with and without Rhodamine B at the indicated age. Females fed similarly on both indicating that Rhodamine B itself does not affect feeding behaviour. Unpaired t-test; n.s.: not significant, p>0.05. 16-20 females/trial. n=5 trials/group. D5(m): 5 days post-emergence, mated; $D2^{PBM}$(m): 2 days post-blood meal, mated; D8(m): 8 days post-emergence, mated. **E**. Determining optimal oviposition timing in *An. stephensi*. Females were tested in groups of 10 and average number of eggs laid per group are shown at the indicated day after the first blood meal. Dashed lines connect the eggs laid by each group on the indicated day. Majority of the eggs were laid 2 days after blood meal. 10 females/group, n=10 replicates/group. $D0^{B}$-$D5^{B}$M: 0-5 days post-blood meal, mated. **F**. Standardisation of the dual choice assay, to determine the choice between blood and sugar in sugar-sated and sugar-starved *An. stephensi* females, co-housed with males. Widefield fluorescent images of females fed on either blood, sugar or both are shown. White arrowhead marks the fluorescence from sugar (spiked with Rhodamine B) in the abdomens of females engorged on blood, indicating that the female has fed on both. This is not observed in the females fed on blood alone. Females fed on sugar alone can be visualised by the fluorescent lean bellies. Unfed females exhibit no fluorescence or engorgement. **G**. Widefield fluorescent images of female *An. stephensi* fed on sugar spiked with Rhodamine B. This setup was used to study the sugar-feeding behaviour of virgins and co-housed females across the reproductive cycle. Addition of Rhodamine B to standard sucrose solution enabled visualisation of even the smallest amounts ingested, as indicated by the white arrowheads, under the fluorescent microscope. **H**. Mating status of sugar-fed and sugar-unfed co-housed females tested in figure 1E, 24-120h post-emergence (D0-D5). n=175-179 females per time point. **I**. Percent participation of virgin and co-housed females tested for host-seeking behaviour in the Y-maze olfactometer (figure 2B), at different stages of the reproductive cycle. All females tested participated well. 17-21 females/trial, n=10 trials/group. Kruskal-Wallis, with Dunn's multiple correction test, p<0.05. Data labelled with different letters are significantly different. (D0: day of emergence ; D5(m): 5 days post-emergence, mated; D5: 5 days post-emergence, virgin; $D3^{PBM}$(m): 3 days post-blood meal, mated; $D3^{PBM}$: 3 days post-blood meal, virgin; $D2^{PO}$(m): 2 days post-oviposition, mated. (m): mated)



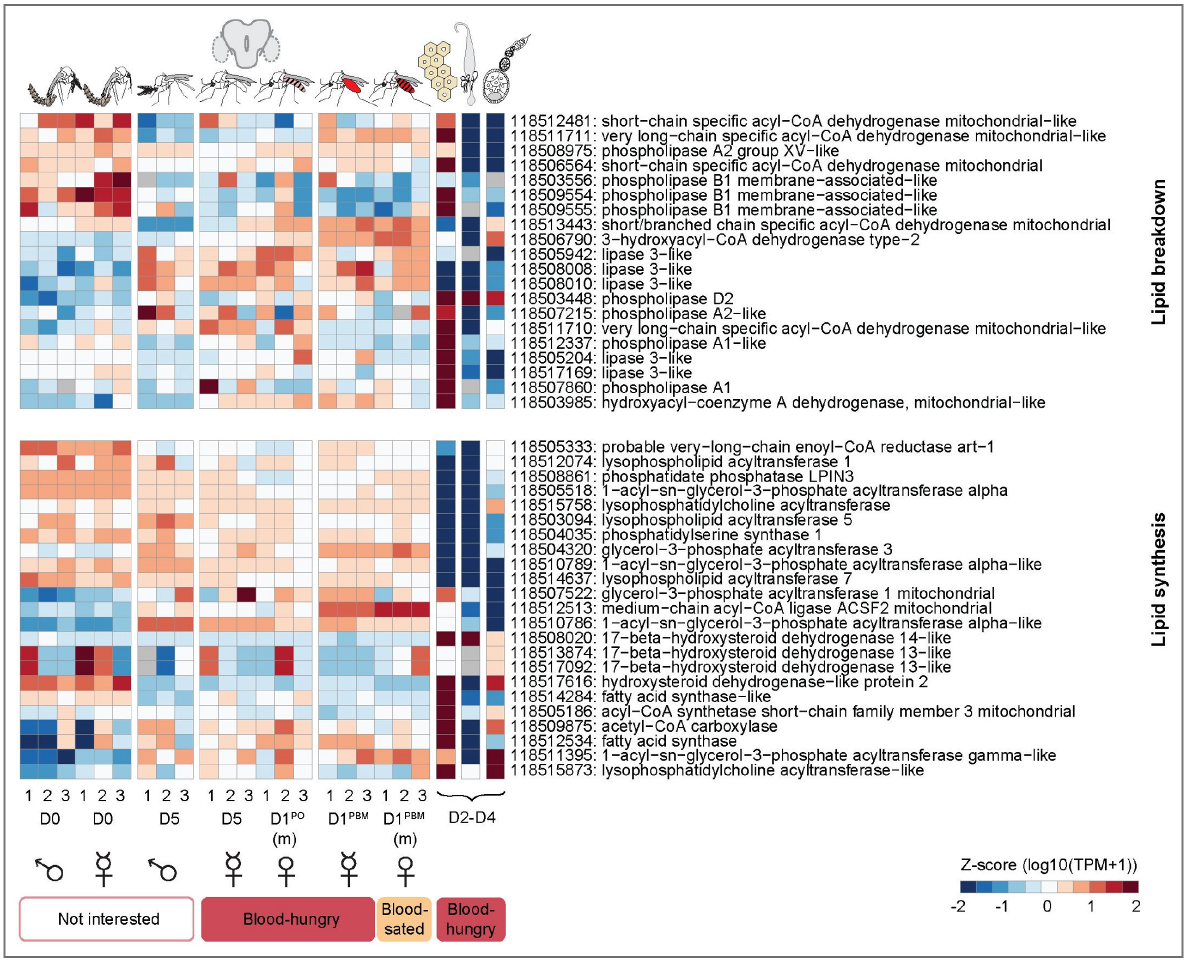
Supplementary Figure 2. Differential expression of genes across various states of blood-hunger and -satiety.

A. Number of genes identified to be differentially up- and downregulated in the central brain of females across the reproductive cycle (indicated at the bottom), compared to either newly emerged males or females (D0), or blood-fed sated females ($D1^{PBM}$(m)). Note that no genes were found to be differentially expressed between mated and virgin blood-fed females, despite their contrasting blood-appetites. **B**. Top 10 Gene Ontology terms for Biological Processes (GOBP), overrepresented in the samples indicated at the bottom, compared to newly emerged males or females (D0). Size of the dot indicates the p-value significance. Processes involved in protein metabolism (highlighted in red) were found to be highly enriched in blood-fed females. (D0: day of emergence; D5: 5 days post-emergence, virgin; $D1^{PBM}$: 1 day post-blood meal, virgin; $D1^{PBM}$(m): 1 day post-blood meal, mated; $D1^{PO}$(m): 1 day post-oviposition, mated. (m): mated)



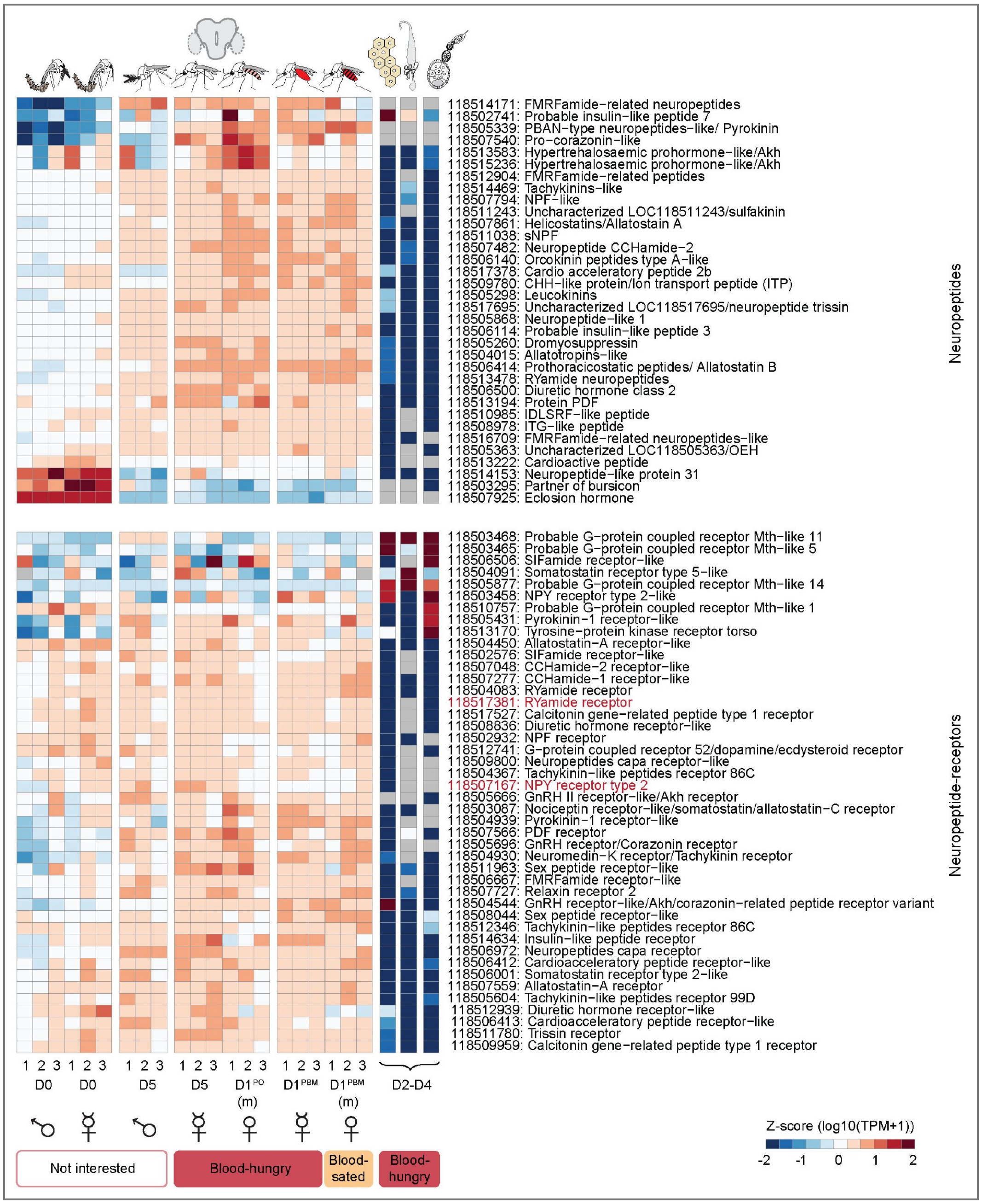
Supplementary Figure 3. Differential expression of genes involved in carbohydrate metabolism across different tissues and states of blood-hunger and -satiety in *An. stephensi*.

Expression patterns of genes (listed on the right) involved in sugar metabolism via different pathways, across the reproductive cycle in the central brain and in the fat body, midgut and ovary of blood-hungry females, are represented as heatmaps. Green box shows downregulation of genes involved in glucose utilisation and the magenta box represents genes involved in glucose storage. Note that both are low in the brains of blood-fed females suggesting that the brain goes into a state of 'sugar rest' after a blood meal. Analysing publicly available data from the midgut, ovary, and fat body of sugar-fed females suggest that energy expenditure via the non-oxidative branch of the pentose phosphate pathway of glucose metabolism is high in these tissues, likely to prime them for the blood meal. RNA-seq data from all central brain samples in triplicates and single sample data for fat body, midgut and ovary (taken from Prasad et al., 2017) are represented as z-score normalised log10 (TPM+1) values. TPM values of zero were excluded from z-score transformation and are displayed as grey cells. (D0: day of emergence; D2-D4: 2-4 days post emergence; D5: 5 days post-emergence, virgin; D1^{PBM}: 1 day post-blood meal, virgin; D1^{PBM}(m): 1 day post-blood meal, mated; D1^{PO} (m): 1 day post-oviposition, mated. (m): mated)



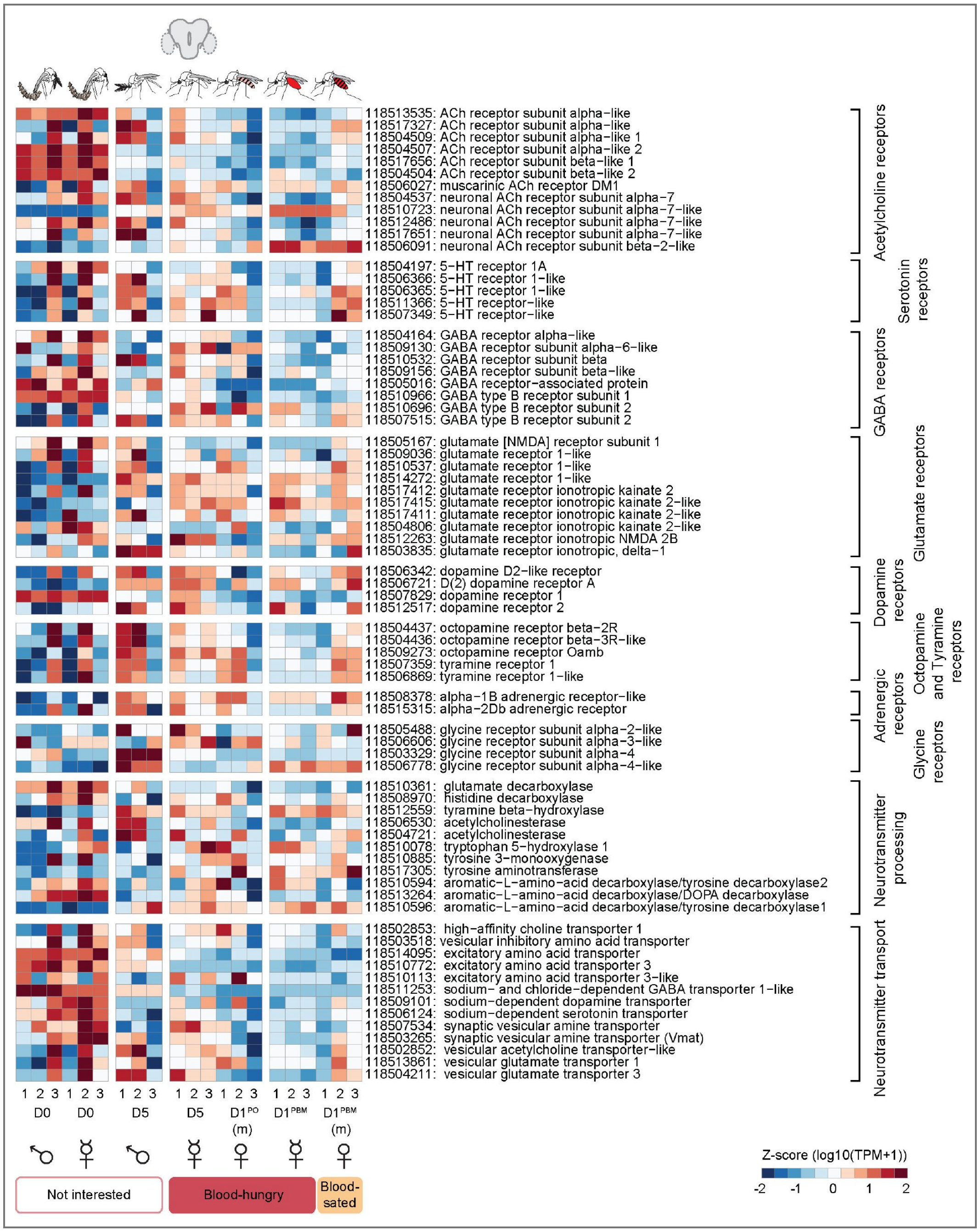
Supplementary Figure 4. **Differential expression of genes involved in lipid metabolism across different tissues and states of blood-hunger and -satiety in *An. stephensi*.**

Expression patterns of genes (listed on the right) involved in lipid metabolism: lipid breakdown (top) and lipid synthesis (bottom), across the reproductive cycle in the central brain and in the fat body, midgut and ovary of blood-hungry females, are represented as heatmaps. No specific patterns were observed in the central brain (pre- and post-blood meal), midgut or ovaries. As expected, fat body showed high expression of genes involved in both the catabolic and anabolic branches of fatty-acid metabolism. RNA-seq data from all central brain samples in triplicates and single sample data for fat body, midgut and ovary (taken from Prasad et al., 2017) are represented as z-score normalised log10 (TPM+1) values. TPM values of zero were excluded from z-score transformation and are displayed as grey cells. (D0: day of emergence; D2-D4: 2-4 days post emergence; D5: 5 days post-emergence, virgin; D1^{PBM}: 1 day post-blood meal, virgin; D1^{PBM}(m): 1 day post-blood meal, mated; D1^{PO}(m): 1 day post-oviposition, mated. (m): mated)



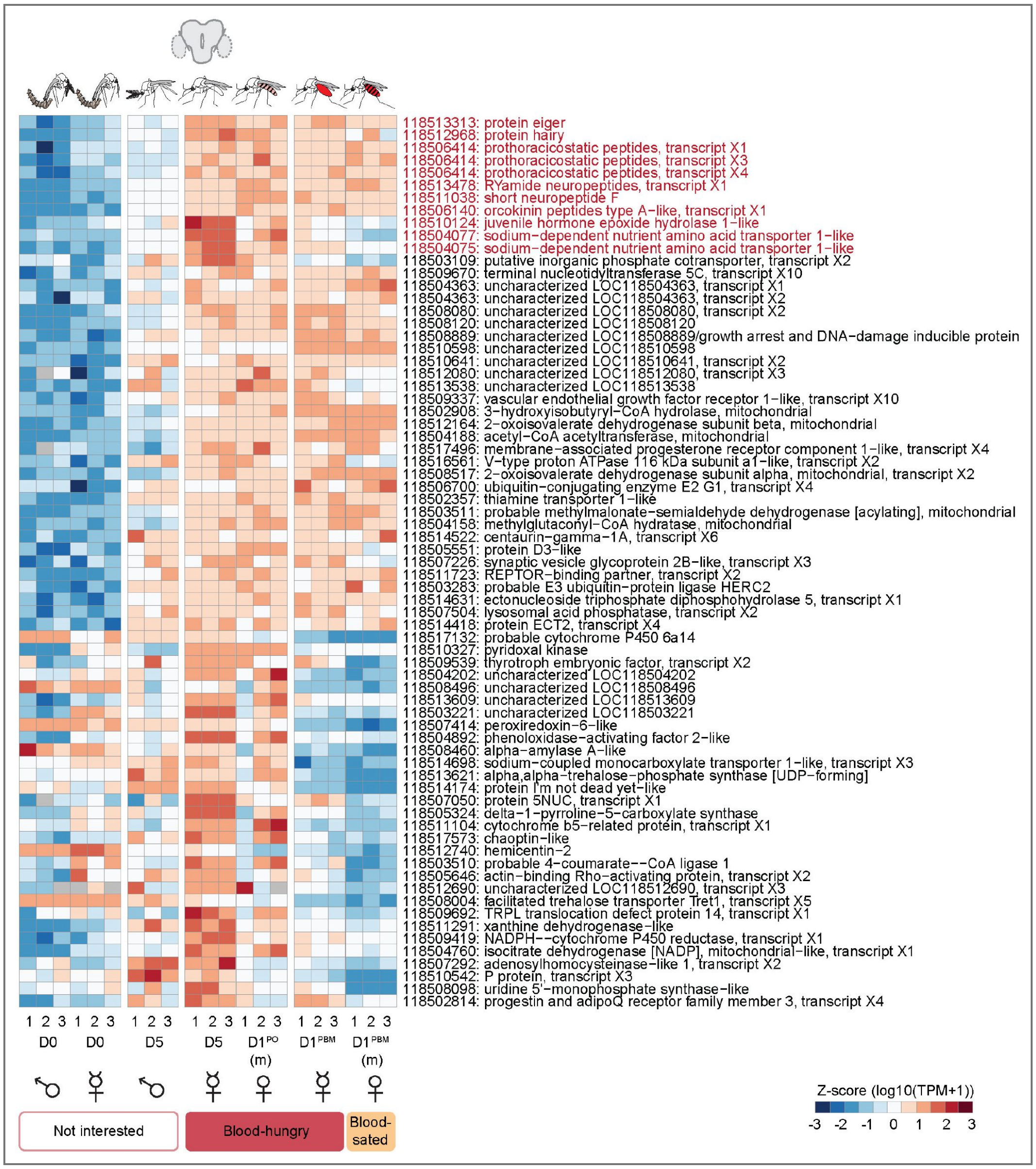
Supplementary Figure 5. **Expression patterns of neuropeptides and neuropeptide receptors across different tissues and states of blood-hunger and -satiety in *An. stephensi*.**

Expression patterns of neuropeptides (top) and neuropeptide receptors (bottom), across the reproductive cycle in the central brain and in the fat body, midgut and ovary of blood-hungry females, are represented as heatmaps. While neuropeptides were observed to be downregulated in the central brain of newly emerged males and females (D0), neuropeptide receptors showed no such patterns. Additionally, both neuropeptides and their receptors were much more abundant in the brain, as compared to fat body, midgut or ovaries. Transcripts for the receptors highlighted in red were visualised *in situ* both in the midgut and brain samples via HCR (figure S10, S11). RNA-seq data from all central brain samples in triplicates and single sample data for fat body, midgut and ovary (taken from Prasad et al., 2017) are represented as z-score normalised log10 (TPM+1) values. TPM values of zero were excluded from z-score transformation and are displayed as grey cells. (D0: day of emergence; D2-D4: 2-4 days post emergence; D5: 5 days post-emergence, virgin; $D1^{PBM}$: 1 day post-blood meal, virgin; $D1^{PBM}$(m): 1 day post-blood meal, mated; $D1^{PO}$(m): 1 day post-oviposition, mated. (m): mated)



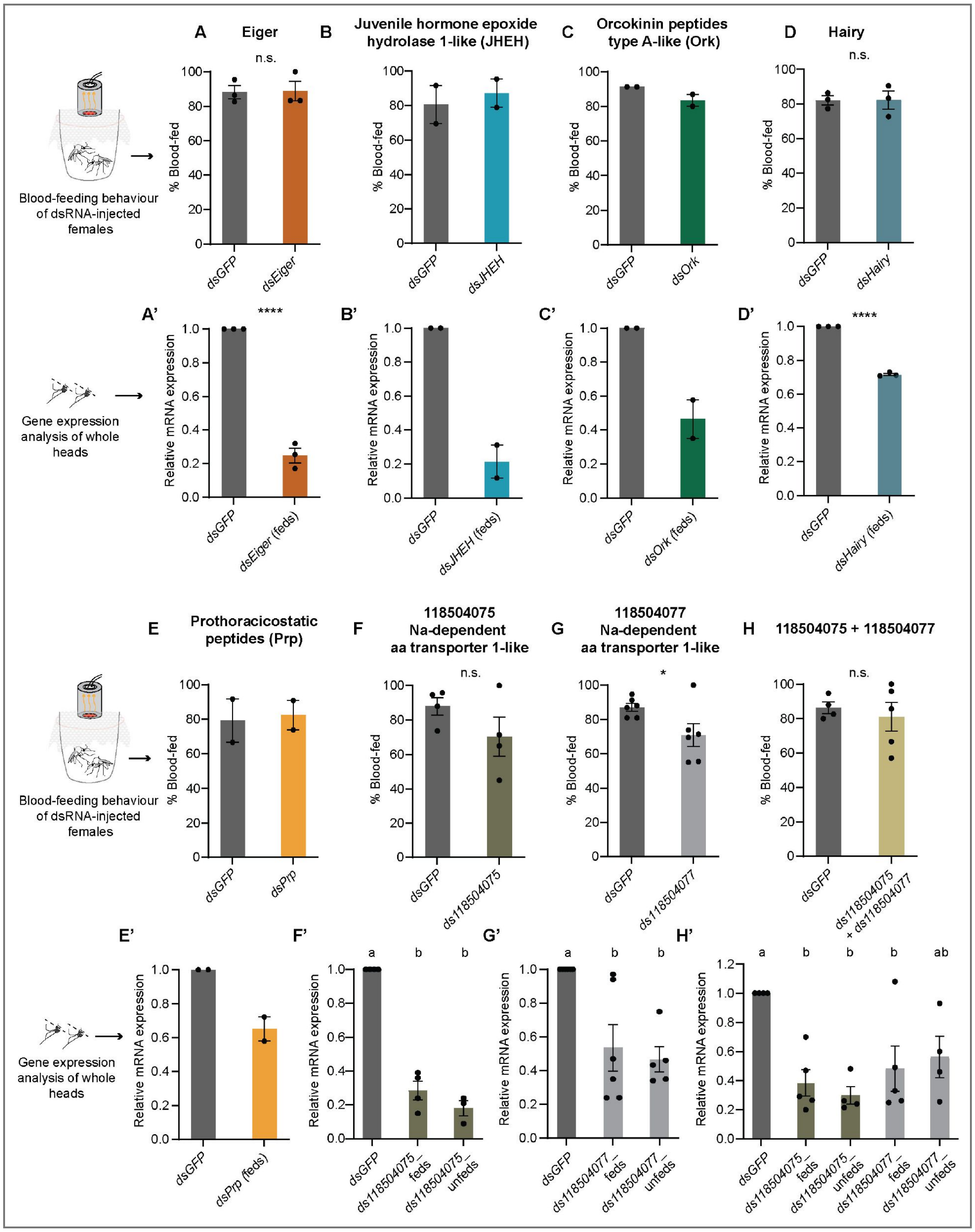
Supplementary Figure 6. **Expression patterns of neurotransmitter-related genes in the central brain of *An. stephensi* across different states of blood-hunger and -satiety.**

Expression patterns of genes (listed on the right) involved in neurotransmitter expression, processing and transport across the reproductive cycle in the central brain, are represented as heatmaps. Interestingly, many of these genes were observed to be upregulated in the newly emerged (D0) mosquitoes as compared to the older and blood-fed mosquitoes. RNA-seq data from all central brain samples in triplicates are represented as z-score normalised log10 (TPM+1) values. (D0: day of emergence; D5: 5 days post-emergence, virgin; D1^{PBM}: 1 day post-blood meal, virgin; D1^{PBM}(m): 1 day post-blood meal, mated; D1^{PO}(m): 1 day post-oviposition, mated. (m): mated)



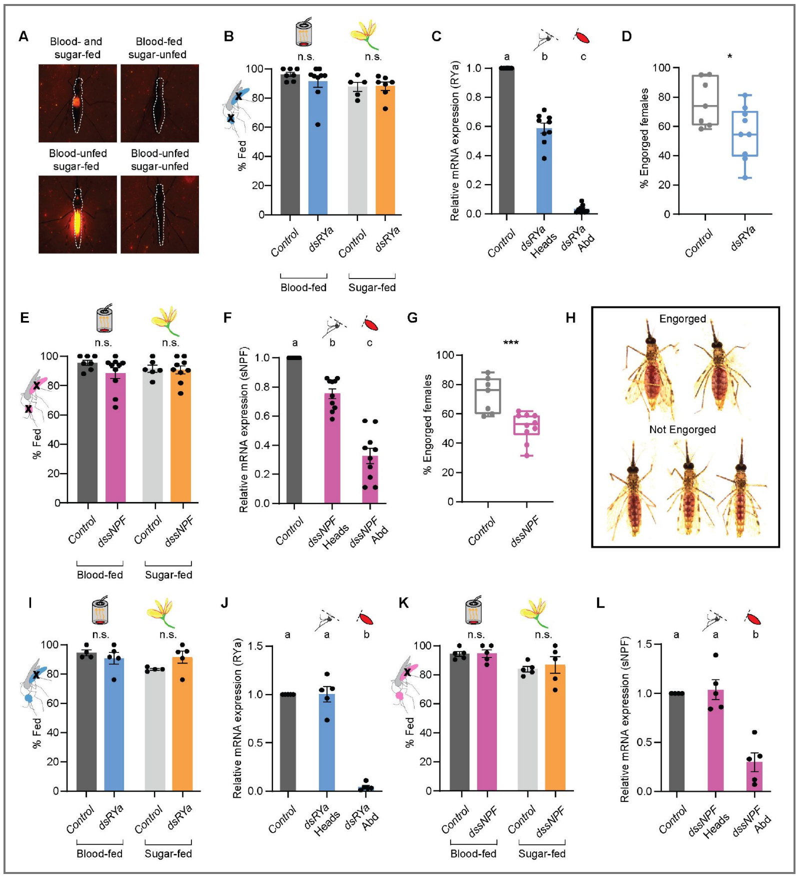
Supplementary Figure 7. **Candidates potentially involved in promoting blood-feeding behaviour in *An. stephensi*.**

71 candidate genes identified as potential promoters of blood-feeding behaviour are listed. These were identified to be commonly upregulated in different conditions of blood-hunger (as compared to conditions of satiety or no-interest), and non-overlapping in males. Nine genes shortlisted for further validation are highlighted in red. RNA-seq data from all central brain samples in triplicates are represented as z-score normalised log10 (TPM+1) values. TPM values of zero were excluded from z-score transformation and are displayed as grey cells. (D0: day of emergence; D5: 5 days post-emergence, virgin; $D1^{PBM}$: 1 day post-blood meal, virgin; $D1^{PBM}$(m): 1 day post-blood meal, mated; $D1^{PO}$(m): 1 day post-oviposition, mated. (m): mated)



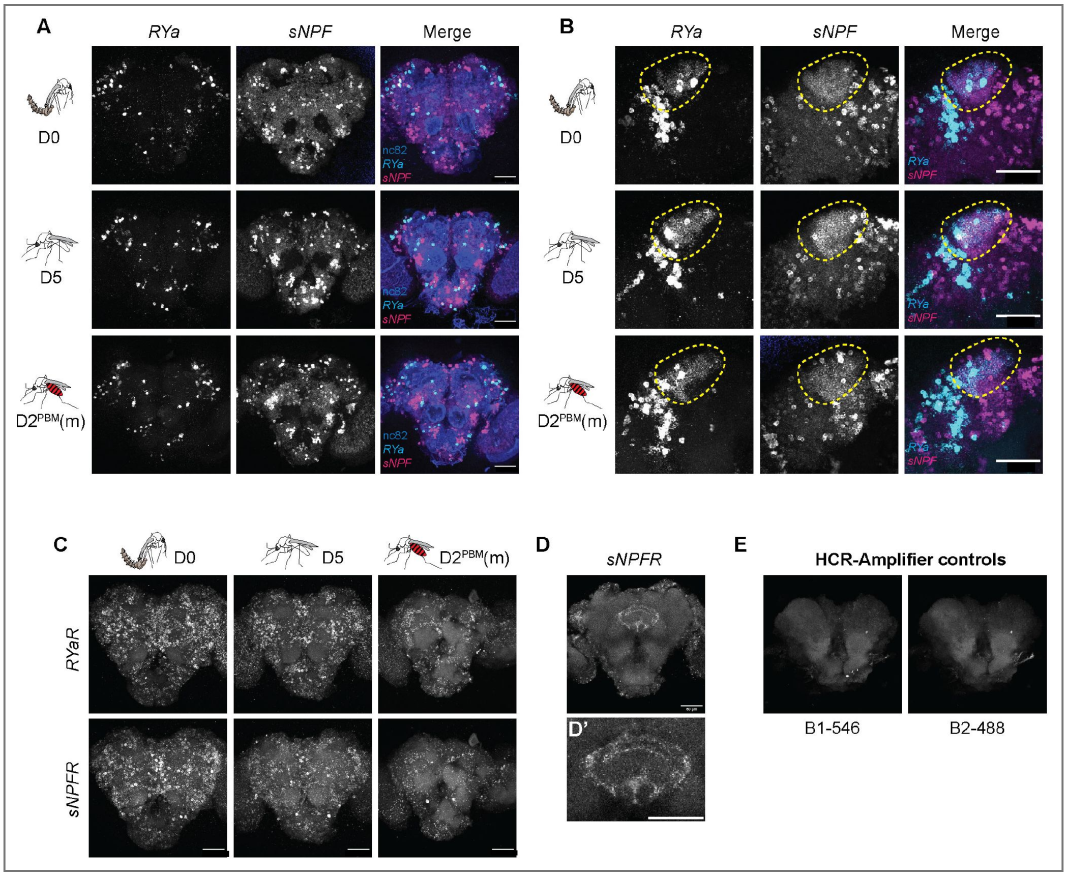
Supplementary Figure 8. Screening of the shortlisted candidates, potentially involved in promoting blood-feeding behaviour in *An. stephensi* via dsRNA-mediated gene knockdown.

A-H. Blood-feeding behaviour of virgin females injected with dsRNA against *Eiger* (A), *Juvenile hormone epoxide hydrolase-1 like (JHEH)* (B), *Orcokinin peptides type-A like (Ork)* (C), *Hairy* (D), *Prothoracicostatic peptides (Prp)* (E), *Na-dependent amino acid transporter-1 like 118504075* (F), *Na-dependent amino acid transporter-1 like 118504077* (G), *118504075+118504077* (H). *dsGFP* was injected as a control in each case. No change in the behaviour was observed for any of the candidates tested. 18-25 females/replicate, n=2-6 replicates/group. Unpaired t-test; n.s.: not significant, $p>0.05$; *$p<0.05$ (A, D, F-H). **A'-H'**. Relative mRNA expression in the heads of virgin *An. stephensi* females injected with the indicated dsRNA, as compared to that in *dsGFP*-injected females, analysed via qPCR. Efficient knockdown could not be achieved for *Hairy* and *Prp*. Unpaired t-test; ****$p<0.0001$ (A', D'). One-way ANOVA with Holm-Šídák multiple comparisons test, $p<0.05$ (F'-H'). Data labelled with different letters are significantly different. For candidates *JHEH* (B), *Ork* (C) and *Prp* (E), significance could not assessed, as statistical testing requires a minimum of three replicates, which were not available for these experiments.



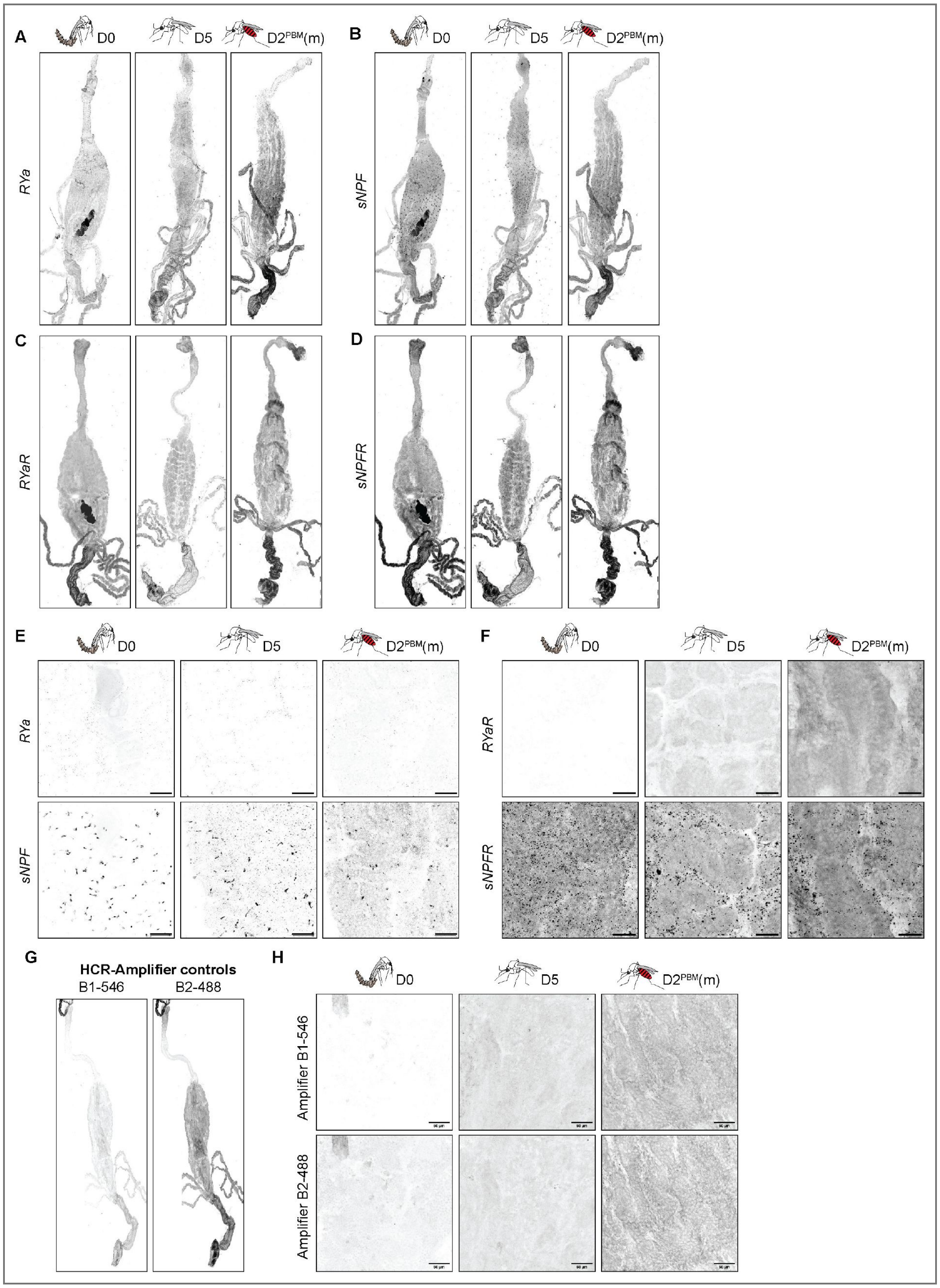
Supplementary Figure 9. **Abdominal knockdown of *RYa* or *sNPF* alone does not contribute to the blood-feeding behaviour in *An. stephensi*.**

A. Widefield fluorescent images of females assayed for both sugar- and blood feeding, post-dsRNA injections. To assay for sugar feeding, injected females were offered Rhodamine B-spiked sugar 24h prior to the behavioural assay. Blood-feeding behaviour was tested as described earlier. Representative images of females fed on both blood and sugar, blood only, sugar only or none, are shown. **B**. Blood- and sugar-feeding behaviour of virgin females where *dsRYa* is knocked down in both heads and abdomens. Controls:*dsGFP*-injected or uninjected females. 16-25 females/replicate, n=5-9 replicates/group. Unpaired t-test; n.s.: not significant, p>0.05. **C**. Relative mRNA expressions of *RYa* in the heads and abdomens of *dsRYa*-injected females assayed in (B). mRNA levels were compared to those in the uninjected or *dsGFP*-injected controls. n=7-9 replicates/group. One-way ANOVA with Holm-Šídák multiple comparisons test, p<0.05. Data labelled with different letters are significantly different. **D**. Percent females engorged on blood upon *dsRYa* knockdown in both heads and abdomens, assayed in (B). Controls:*dsGFP*-injected or uninjected females. n=7-9 replicates/group. Unpaired t-test; *p<0.05. **E**. Blood- and sugar-feeding behaviour of virgin females where *sNPF* is knocked down in both heads and abdomens. Controls:*dsGFP*-injected or uninjected females. 17-26 females/replicate, n=6-10 replicates/group. Unpaired t-test; n.s.: not significant, p>0.05. **F**. Relative mRNA expressions of *sNPF* in the heads and abdomens of *sNPF* - injected females assayed in (E). mRNA levels were compared to those in the uninjected or *dsGFP*-injected controls. n=7-10 replicates/group. One-way ANOVA with Holm-Šídák multiple comparisons test, p<0.05. Data labelled with different letters are significantly different. **G**. Percent females engorged on blood upon *sNPF* knockdown in both heads and abdomens, assayed in (E). Controls:*dsGFP*-injected or uninjected females. n=7-10 replicates/group. Unpaired t-test; ***p<0.001. **H**. Representative widefield images of females considered to be engorged or not in (D) and (G). **I**. Blood- and sugar-feeding behaviour of virgin females where *dsRYa* is knocked down in abdomens only. Controls:*dsGFP*-injected females. 19-24 females/replicate, n=4-5 replicates/group. Unpaired t-test; n.s.: not significant, p>0.05. **J**. Relative mRNA expressions of *RYa* in the heads and abdomens of *dsRYa*-injected females assayed in (I). mRNA levels were compared to those *dsGFP*-injected controls. n=4-5 replicates/group. One-way ANOVA with Holm-Šídák multiple comparisons test, p<0.05. Data labelled with different letters are significantly different. **K**. Blood- and sugar-feeding behaviour of virgin females where *sNPF* is knocked down in abdomens only. Controls:*dsGFP*-injected females. 19-23 females/replicate, n=5 replicates/group. Unpaired t-test; n.s.: not significant, p>0.05. **L**. Relative mRNA expressions of *sNPF* in the heads and abdomens of *sNPF* - injected females assayed in (K). mRNA levels were compared to those *dsGFP*-injected controls. n=5 replicates/group. One-way ANOVA with Holm-Šídák multiple comparisons test, p<0.05. Data labelled with different letters are significantly different.



Supplementary Figure 10. Transcripts of neuropeptides *sNPF, RYa* and their receptors are expressed in *An. stephensi* central brain.

A, B. *RYa* and *sNPF* mRNA expression in the central brain of females uninterested in blood (D0), blood-hungry females (D5) and blood-sated females (D2^{PBM}(m)), analysed via HCR. Transcripts of both neuropeptides are expressed in several clusters across the central brain (A). While the clusters are largely non-overlapping, co-expression was observed in the mushroom body Kenyon cells, indicated by the marked area (B). Maximum intensity projections of all confocal stacks (A) or selected stacks (B) are shown. Scale bar, 50µm. **C**. *RYa receptor (RYaR)* and *sNPF receptor (sNPFR)* mRNA expression in the central brain of females uninterested in blood (D0), blood-hungry females (D5) and blood-sated (D2^{PBM}(m)), analysed via HCR. Transcripts of both receptors were found to be expressed in multiple cells. Maximum intensity projections of confocal stacks are shown. Scale bar, 50µm. **D**. *sNPFR* mRNA expression in the central complex. A magnified image is shown in D'. Maximum intensity projection of selected confocal stacks is shown. Scale bar, 50µm. **E**. Central brain samples incubated with HCR-Amplifiers only, to serve as a control for background signal in the absence of HCR probes. No signal was observed for either of the amplifiers used (B1 labelled with 546 fluorophore (B1-546) and B2 labelled with 488 fluorophore (B2-488)). (D0: day of emergence; D5: 5 days post-emergence, virgin; D2^{PBM}(m): 2 days post-blood meal, mated. (m): mated)



Supplementary Figure 11. **Transcripts of *sNPF* and *sNPFR* are expressed in *An. stephensi* posterior midgut.**

A-F. *RYa* (A), *sNPF* (B), *RYaR* (C) and *sNPFR* (D) mRNA expression in the guts of females uninterested in blood (D0), blood-hungry females (D5) and blood-sated females (D2^{PBM}(m)), analysed via HCR. Transcripts of both *sNPF* and its receptor *sNPFR* are expressed in the posterior midgut, at all stages analysed. No expression of *RYa* or *RYaR* was observed. Magnified images of transcript expression of neuropeptides and their receptors are shown in E and F, respectively. Scale bar, 50μm. **G**,**H**. Gut samples incubated with HCR-Amplifiers only, to serve as a control for background signal in the absence of HCR probes. No signal was observed for either of the amplifiers used (B1 labelled with 546 fluorophore (B1-546) and B2 labelled with 488 fluorophore (B2-488)). Scale bar, 50μm. (D0: day of emergence; D5: 5 days post-emergence, virgin; D2^{PBM}(m): 2 days post-blood meal, mated. (m): mated)

S.No.	Primer name	Primer Sequence
1.	T7-GFP_F_RNAi	TAATACGACTCACTATAGGGAGACTGGTCGAGCTGGACGGCGA
2.	T7-GFP_R_RNAi	TAATACGACTCACTATAGGGAGACTTCTCGTTGGGGTCTTTGCTCAGG
3.	T7-RYamide_F_RNAi	TAATACGACTCACTATAGGGATGACCTGGCGAACGATGAAAC
4.	T7-RYamide_R_RNAi	TAATACGACTCACTATAGGGGGTCGCCATTCTCGTTGAACTGG
5.	RYamide_qPCR_F	ACCGATGCTACAGTCGTGATTCG
6.	RYamide_qPCR_R	TAGGCTAGTTGCATGGATCGTACC
7.	T7-sNPF_F_RNAi	TAATACGACTCACTATAGGGTTTGATGTCGGAGTCACTGCATCC
8.	T7-sNPF_R_RNAi	TAATACGACTCACTATAGGGTCACTCGCATCGTTAACCAACTGC
9.	sNPF_qPCR_F	CAATGAGCATCAGCTAGCACCG
10.	sNPF_qPCR_R	TCTAGCGCTTTATCCTCGGAGG
11.	T7-Hairy_RNAi_F	TAATACGACTCACTATAGGGCTCGTATCAACAACTGTCTGAACG
12.	T7-Hairy_RNAi_R	TAATACGACTCACTATAGGGGTACTTGGGTGGTGAGCGTTTGG
13.	Hairy_qPCR_F	AATCGTCGGAGCAACAAACCGATC
14.	Hairy_qPCR_R	GATGCTTCACCGTCATCTCCAGG
15.	T7-Eiger_RNAi_F	TAATACGACTCACTATAGGGCTGGACGATAGCGAGGAAAAGG
16.	T7-Eiger_RNAi_R	TAATACGACTCACTATAGGGGCTCATTCGGTTCAGTATTTGTGC
17.	Eiger_qPCR_F	GGGCGTAAGACATCACTCACGC
18.	Eiger_qPCR_R	TACGATTTCCTTCTGCGTTCCAGG
19.	T7-JHEH_RNAi_F	TAATACGACTCACTATAGGGGATTATTGGGGACCCGGAAATGG
20.	T7-JHEH_RNAi_R	TAATACGACTCACTATAGGGCGTTTGTGACCGATGCGTTCC
21.	JHEH_qPCR_F	ATCGCCAGTGCCAGTTGAGC
22.	JHEH_qPCR_R	TTGCTCCATTTCCGGGTCCC
23.	T7-Prp_RNAi_F	TAATACGACTCACTATAGGGAATGTATCGCGGTCTGCTTTGC
24.	T7-Prp_RNAi_R	TAATACGACTCACTATAGGGCGACTAATTTCCTGCCGTTTGCC
25.	Prp_qPCR_F	AAGTTCGGTGCGGCCTGG
26.	Prp_qPCR_R	TCTGCTGCTGCTGCTGCTG
27.	T7-4075_RNAi_F	TAATACGACTCACTATAGGGGATACGGTCACTTCGCTGGTGG
28.	T7-4075_RNAi_R	TAATACGACTCACTATAGGGATGAACAGTACGGTGATGGGCG
29.	4075_qPCR_F	CCTTCGCCTGGATCTATGGCG
30.	4075_qPCR_R	CGAGCGCATTGTAACCGAGTGG
31.	T7-4077_RNAi_F	TAATACGACTCACTATAGGGCTCGACACATTCACCTCCATCG
32.	T7-4077_RNAi_R	TAATACGACTCACTATAGGGATAAGAATGAGCAGTGTGGCGG
33.	4077_qPCR_F	ATCTACGGCGTGGATCGTATCTGC
34.	4077_qPCR_R	CCTAAAACGTTGTAGCCGATCGGG
35.	T7-Orcokinin_F_RNAi	TAATACGACTCACTATAGGGGTCTGTGCCGTACTGCTGTTCG
36.	T7-Orcokinin_R_RNAi	TAATACGACTCACTATAGGGGATCGTGCTTCAGGTTGTTACCGG
37.	Orcokinin_qPCR_F	GCTGAACGATGGCCAACTAAAGC
38.	Orcokinin_qPCR_R	CTTGTCGTACATCGGTGCCAATCC
39.	RpS7_qPCR_F	AGGTTGTCGGCAAGCGTATCC
40.	RpS7_qPCR_R	AGCTTCTTGTACACCGACGCG

Table 1. **List of primers used in the study for *dsRNA*-mediated gene knockdown and quantitative PCR (qPCR)**

Data availability

Raw fastq files have been deposited at the National Center for Biotechnology Information (NCBI) Sequence Read Archive (SRA), under the BioProject ID PRJNA1297931.

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Additional information

Author contributions

PB conceptualised the project, performed all experiments except RNA-seq and host-seeking assay, performed all analysis, including RNA-seq, wrote and reviewed the manuscript. RP performed the RNA-seq experiments and analysis, standardised HCR, and reviewed the manuscript. PDB performed the host-seeking assay, provided technical assistance during HCR standardisations and determination of mating status, and reviewed the manuscript. SQS conceptualised and supervised the project and wrote and reviewed the manuscript.

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Additional files

Supplementary sheet 1. Summary of statistical analysis of feeding behaviours using Generalised Linear Mixed Models (GLMM).

Supplementary sheet 2. Table of transcript abundance (in TPM values) for genes identified in the central brain RNAseq data across the gonotrophic cycle.

Supplementary sheet 3. Table of differentially expressed genes (DEGs) identified across states of blood-hunger and -satiety.

Supplementary movie. *sNPF* expression in the central brain of females uninterested in blood (D0), blood-hungry females (D5) and blood-sated females ($D2^{PBM}$(m)). Confocal stacks of whole brains probed for *sNPF* mRNA expression via HCR *in situ* hybridization, at three different conditions of blood-hunger and -satiety, are shown as movies. Note the appearance of novel clusters expressing *sNPF* transcripts (indicated by magenta arrows) in the subesophageal ganglionic region of the brain, only in the blood-hungry (D5) condition.

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Peer reviews

Reviewer #1 (Public review):

Summary:

Here Bansal et al., present a study on the fundamental blood and nectar feeding behaviors of the critical disease vector, Anopheles stephensi. The study encompasses not just the fundamental changes in blood feeding behaviors of the crucially understudied vector, but then use a transcriptomic approach to identify candidate neuromodulation path ways which influence blood feeding behavior in this mosquito species. The authors then provide evidence through RNAi knockdown of candidate pathways that the neuromodulators sNPF and Rya modulate feeding either via their physiological activity in the brain alone or through joint physiological activity along the brain-gut axis (but critically not the gut alone). Overall, I found this study to be built on tractable, well-designed behavioral experiments.

Their study begins with a well-structured experiment to assess how the feeding behaviors of A. stephensi changes over the course of its life history and in response to its age, mating and oviposition status. The authors are careful and validate their experimental paradigm in the more well-studied Ae. aegypti, and are able to recapitulate the results of prior studies which show that mating is pre-requisite for blood feeding behaviors in Ae. aegypt. Here they find A. stephensi like another Anopheline mosquitoes has a more nuanced regulation of its blood and nectar feeding behaviors.

The authors then go on to show in a Y- maze olfactometer that to some degree, changes in blood feeding status depend on behavioral modulation to host-cues, and this is not likely to be a simple change to the biting behaviors alone. I was especially struck by the swap in valence of the host-cues for the blood-fed and mated individuals which had not yet oviposited. This indicates that there is a change in behavior that is not simply desensitization to host-cues while navigating in flight, but something much more exciting happening.

The authors then use a transcriptomic approach to identify candidate genes in the blood feeding stages of the mosquito's life cycle to identify a list of 9 candidates which have a role in regulating the host-seeking status of A. stephensi. Then through investigations of gene knockdown of candidates they identify the dual action of RYa and sNPF and candidate neuromodulators of host-seeking in this species. Overall, I found the experiments to be well-designed. I found the molecular approach to be sound. While I do not think the molecular approach is necessarily an all-encompassing mechanism identification (owing mostly to the fact that genetic resources are not yet available in A. stephensi as they are in other dipteran models), I think it sets up a rich lines of research questions for the neurobiology of mosquito behavioral plasticity and comparative evolution of neuromodulator action.

Strengths:

I am especially impressed by the authors' attention to small details in the course of this article. As I read and evaluated this article I continued to think how many crucial details I may have missed if I were the scientist conducting these experiments. That attention to detail paid off in spades and allowed the authors to carefully tease apart molecular candidates of blood-seeking stages. The authors top down approach to identifying RYamide and sNPF starting from first principles behavioral experiments is especially comprehensive. The results from both the behavioral and molecular target studies will have broad implications for the vectorial capacity of this species and comparative evolution of neural circuit modulation.

I believe the authors have adequately addressed all of my concerns; however, I think an accompanying figure to match the explained methods of the tissue-specific knockdown would help readers. The methods are now explicitly written for the timing and concentrations required to achieve tissue-specific knockdown, but seeing the data as a supplement would be especially reassuring given the critical nature of tissue-specific knockdown to the final interpretations of this paper.

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Reviewer #2 (Public review):

Summary:

In this manuscript, Bansal et al examine and characterize feeding behaviour in Anopheles stephensi mosquitoes. While sharing some similarities to the well-studied Aedes aegypti mosquito, the authors demonstrate that mated-females, but not unmated (virgin) females, exhibit suppression in their blood-feeding behaviour. Using brain transcriptomic analysis comparing sugar fed, blood fed and starved mosquitoes, several candidate genes potentially responsible for influencing blood-feeding behaviour were identified, including two neuropeptides (short NPF and RYamide) that are known to modulate feeding behaviour in other mosquito species. Using molecular tools including in situ hybridization, the authors map the distribution of cells producing these neuropeptides in the nervous system and in the gut. Further, by implementing systemic RNA interference (RNAi), the study suggests that both neuropeptides appear to promote blood-feeding (but do not impact sugar feeding) although the impact was observed only after both neuropeptide genes underwent knockdown.

While the authors have addressed most of the concerns of the original manuscript, a few issues remain. Particularly, the following two points:

(5) Figure 4

The authors state that there is more efficient knockdown in the head of unfed females; however, this is not accurate since they only get knockdown in unfed animals, and no evidence of any knockdown in fed animals (panel D). This point should be revised in the results test as well.

Perhaps we do not understand the reviewer's point or there has been a misunderstanding. In Figure 4D, we show that while there is more robust gene knockdown in unfed females, blood-fed females also showed modest but measurable knockdowns ranging from 5-40% for RYamide and 2-21% for sNPF.

NEW-

In both the dsRNA treatments where animals were fed, neither was significantly different from control. Therefore, there is no change, and indeed this is confirmed by the author's labelling of the figure stats in panel 4D.

In addition, do the uninjected and dsGFP-injected relative mRNA expression data reflect combined RYa and sNPF levels? Why is there no variation in these data,...

In these qPCRs, we calculated relative mRNA expression using the delta-delta Ct method (see line 975). For each neuropeptide its respective control was used. For simplicity, we combined the RYa and sNPF control data into a single representation. The value of this control is invariant because this method sets the control baseline to a value of 1.

NEW-

The authors are claiming that there is no variation between individual qPCR experiments (particularly in their controls)? Normally, one uses a known standard value (or calibrator) across multiple experiments/plates so that variation across biological replicates can be assessed. This has an impact on statistical analyses since there is no variation in the control data. Indeed, this impacts all figures/datasets in the manuscript where qPCR data is presented. All the controls have zero variation!

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Reviewer #3 (Public review):

Summary:

This manuscript investigates the regulation of host-seeking behavior in Anopheles stephensi females across different life stages and mating states. Through transcriptomic profiling, the authors identify differential gene expression between "blood-hungry" and "blood-sated" states. Two neuropeptides, sNPF and RYamide, are highlighted as potential mediators of host-seeking behavior. RNAi knockdown of these peptides alters host-seeking activity, and their expression is anatomically mapped in the mosquito brain (sNPF and RYamide) and midgut (sNPF only).

Strengths:

(1) The study addresses an important question in mosquito biology, with relevance to vector control and disease transmission.

(2) Transcriptomic profiling is used to uncover gene expression changes linked to behavioral states.

(3) The identification of sNPF and RYamide as candidate regulators provides a clear focus for downstream mechanistic work.

(3) RNAi experiments demonstrate that these neuropeptides are necessary for normal host-seeking behavior.

(4) Anatomical localization of neuropeptide expression adds depth to the functional findings.

Weaknesses:

(1) The title implies that the neuropeptides promote host-seeking, but sufficiency is not demonstrated and some conclusions appear premature based on the current data. The support for this conclusion would be strengthened with functional validation using peptide injection or genetic manipulation.

(2) The identification of candidate receptors is promising, but the manuscript would be significantly strengthened by testing whether receptor knockdowns phenocopy peptide knockdowns. Without this, it is difficult to conclude that the identified receptors mediate the behavioral effects.

(3) Some important caveats, such as variation in knockdown efficiency and the possibility of off-target effects, are not adequately discussed.

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Author response:

The following is the authors' response to the original reviews.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *Bansal et al. present a study on the fundamental blood and nectar feeding behaviors of the critical disease vector, Anopheles stephensi. The study encompasses not just the fundamental changes in blood feeding behaviors of the crucially understudied vector, but then uses a transcriptomic approach to identify candidate neuromodulation pathways which influence blood feeding behavior in this mosquito species. The authors then provide evidence through RNAi knockdown of candidate pathways that the neuromodulators sNPF and Rya modulate feeding either via their physiological activity in the brain alone or through joint physiological activity along the brain-gut axis (but critically not the gut alone). Overall, I found this study to be built on tractable, well-designed behavioral experiments.*
>
> *Their study begins with a well-structured experiment to assess how the feeding behaviors of A. stephensi change over the course of its life history and in response to its age, mating, and oviposition status. The authors are careful and validate their experimental paradigm in the more well-studied Ae. aegypti, and are able to recapitulate the results of prior studies, which show that mating is a prerequisite for blood feeding behaviors in Ae. aegypt. Here they find A. Stephensi, like other Anopheline mosquitoes, has a more nuanced regulation of its blood and nectar feeding behaviors.*
>
> *The authors then go on to show in a Y-maze olfactometer that ,to some degree, changes in blood feeding status depend on behavioral modulation to host cues, and this is not likely to be a simple change to the biting behaviors alone. I was especially struck by the swap in valence of the host cues for the blood-fed and mated individuals, which had not yet oviposited. This indicates that there is a change in behavior that is not simply*

> *desensitization to host cues while navigating in flight, but something much more exciting is happening.*
>
> *The authors then use a transcriptomic approach to identify candidate genes in the blood-feeding stages of the mosquito's life cycle to identify a list of 9 candidates that have a role in regulating the host-seeking status of A. stephensi. Then, through investigations of gene knockdown of candidates, they identify the dual action of RYa and sNPF and candidate neuromodulators of host-seeking in this species. Overall, I found the experiments to be well-designed. I found the molecular approach to be sound. While I do not think the molecular approach is necessarily an all-encompassing mechanism identification (owing mostly to the fact that genetic resources are not yet available in A. stephensi as they are in other dipteran models), I think it sets up a rich line of research questions for the neurobiology of mosquito behavioral plasticity and comparative evolution of neuromodulator action.*

We appreciate the reviewer's detailed summary of our work. We thank them for their positive comments and agree with them on the shortcomings of our approach.

> *Strengths:*
>
> *I am especially impressed by the authors' attention to small details in the course of this article. As I read and evaluated this article, I continued to think about how many crucial details could potentially have been missed if this had not been the approach. The attention to detail paid off in spades and allowed the authors to carefully tease apart molecular candidates of blood-seeking stages. The authors' top-down approach to identifying RYamide and sNPF starting from first principles behavioral experiments is especially comprehensive. The results from both the behavioral and molecular target studies will have broad implications for the vectorial capacity of this species and comparative evolution of neural circuit modulation.*

We really appreciate that the reviewer has recognised the attention to detail we have tried to put, thank you!

> *Weaknesses:*
>
> *There are a few elements of data visualizations and methodological reporting that I found confusing on a first few read-throughs. Figure 1F, for example, was initially confusing as it made it seem as though there were multiple 2-choice assays for each of the conditions. I would recommend removing the "X" marker from the x-axis to indicate the mosquitoes did not feed from either nectar, blood, or neither in order to make it clear that there was one assay in which mosquitoes had access to both food sources, and the data quantify if they took both meals, one meal, or no meals.*

We thank the reviewer for flagging the schematic in figure 1F. As suggested, we have removed the "X" markers from the x-axis and revised the axis label from "choice of food" to "choice made" to better reflect what food the mosquitoes chose in the assay. For clarity, we have now also plotted the same data as stacked graphs at the bottom of Fig. 1F, which clearly shows the proportion of mosquitoes fed on each particular choice. We avoid the stacked graph as the sole representation of this data, as it does not capture the variability in the data.

> *I would also like to know more about how the authors achieved tissue-specific knockdown for RNAi experiments. I think this is an intriguing methodology, but I could not figure out from the methods why injections either had whole-body or abdomen-specific knockdown.*

The tissue-specific knockdown (abdomen only or abdomen+head) emerged from initial standardisations where we were unable to achieve knockdown in the head unless we used higher concentrations of dsRNA and did the injections in older females. We realised that this gave us the opportunity to isolate the neuronal contribution of these neuropeptides in the phenotype produced. Further optimisations revealed that injecting dsRNA into 0-10h old females produced abdomen-specific knockdowns without affecting head expression, whereas injections into 4 days old females resulted in knockdowns in both tissues. Moreover, head knockdowns in older females required higher dsRNA concentrations, with knockdown efficiency correlating with the amount injected. In contrast, abdominal knockdowns in younger females could be achieved even with lower dsRNA amounts.

We have mentioned the knockdown conditions- time of injection and the amount dsRNA injected- for tissue-specific knockdowns in methods but realise now that it does not explain this well enough. We have now edited it to state our methodology more clearly (see lines 932-948).

> *I also found some interpretations of the transcriptomic to be overly broad for what transcriptomes can actually tell us about the organism's state. For example, the authors mention, "Interestingly, we found that after a blood meal, glucose is neither spent nor stored, and that the female brain goes into a state of metabolic 'sugar rest', while actively processing proteins (Figure S2B, S3)".*
>
> *This would require a physiological measurement to actually know. It certainly suggests that there are changes in carbohydrate metabolism, but there are too many alternative interpretations to make this broad claim from transcriptomic data alone.*

We thank the reviewer for pointing this out and agree with them. We have now edited our statement to read:

"Instead, our data suggests altered carbohydrate metabolism after a blood meal, with the female brain potentially entering a state of metabolic 'sugar rest' while actively processing proteins (Figure S2B, Sti). However, physiological measurements of carbohydrate and protein metabolism will be required to confirm whether glucose is indeed neither spent nor stored during this period." See lines 271-277.

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *In this manuscript, Bansal et al examine and characterize feeding behaviour in Anopheles stephensi mosquitoes. While sharing some similarities to the well-studied Aedes aegypti mosquito, the authors demonstrate that mated females, but not unmated (virgin) females, exhibit suppression in their bloodfeeding behaviour. Using brain transcriptomic analysis comparing sugar-fed, blood-fed, and starved mosquitoes, several candidate genes potentially responsible for influencing blood-feeding behaviour were identified, including two neuropeptides (short NPF and RYamide) that are known to modulate feeding behaviour in other mosquito species. Using molecular tools, including in situ hybridization, the authors map the distribution of cells producing these neuropeptides in the nervous system and in the gut. Further, by implementing systemic RNA interference (RNAi), the study suggests that both neuropeptides appear to promote blood-feeding (but do not impact sugar feeding), although the impact was observed only after both neuropeptide genes underwent knockdown.*
>
> *Strengths and/or weaknesses:*
>
> *Overall, the manuscript was well-written; however, the authors should review carefully, as some sections would benefit from restructuring to improve clarity. Some statements need to be rectified as they are factually inaccurate.*

> *Below are specific concerns and clarifications needed in the opinion of this reviewer:*
>
> *(1) What does "central brains" refer to in abstract and in other sections of the manuscript (including methods and results)? This term is ambiguous, and the authors should more clearly define what specific components of the central nervous system was/were used in their study.*

Central brain, or mid brain, is a commonly used term to refer to brain structures/neuropils without the optic lobes (For example: *https://www.nature.com/articles/s41586-024-07686-5*). In this study we have focused our analysis on the central brain circuits involved in modulating blood-feeding behaviour and have therefore excluded the optic lobes. As optic lobes account for nearly half of all the neurons in the mosquito brain (*https://pmc.ncbi.nlm.nih.gov/articles/PMC8121336/*), including them would have disproportionately skewed our transcriptomic data toward visual processing pathways.

We have indicated this in figure 3A and in the methods (see lines 800-801, 812). We have now also clarified it in the results section for neurotranscriptomics to avoid confusion (see lines 236-237).

> *(2) The abstract states that two neuropeptides, sNPF and RYamide are working together, but no evidence is summarized for the latter in this section.*

We thank the reviewer for pointing this out. We have now added a statement "This occurs in the context of the action of RYa in the brain" to end of the abstract, for a complete summary of our proposed model.

> *(3) Figure 1*
>
> *Panel A: This should include mating events in the reproductive cycle to demonstrate differences in the feeding behavior of Ae. aegypti.*

Our data suggest that mating can occur at any time between eclosion and oviposition in An. stephensi and between eclosion and blood feeding in Ae. aegypti. Adding these into (already busy) 1A, would cloud the purpose of the schematic, which is to indicate the time points used in the behavioural assays and transcriptomics.

> *Panel F: In treatments where insects were not provided either blood or sugar, how is it that some females and males had fed? Also, it is unclear why the y-axis label is % fed when the caption indicates this is a choice assay. Also, it is interesting that sugar-starved females did not increase sugar intake. Is there any explanation for this (was it expected)?*

We apologise for the confusion. The experiment is indeed a choice assay in which sugar-starved or sugar-sated females, co-housed with males, were provided simultaneous access to both blood and sugar, and were assessed for the choice made (indicated on the x-axis): both blood and sugar, blood only, sugar only, or neither. The x-axis indicates the choice made by the mosquitoes, not the choice provided in the assay, and the y-axis indicates the percentage of males or females that made each particular choice. We have now removed the "X" markers from the x-axis and revised the axis label from "choice of food" to "choice made" to better reflect what food the mosquitoes chose to take.

In this assay, we scored females only for the presence or absence of each meal type (blood or sugar) and are therefore unable to comment on whether sugar-starved females consumed more sugar than sugarsated females. However, when sugar-starved, a higher proportion of females consumed both blood and sugar, while fewer fed on blood alone.

For clarity, we have now also plotted the same data as stacked graphs at the bottom of Fig. 1F, which clearly shows the proportion of mosquitoes fed on each particular choice. We avoid the stacked graph as the sole representation of this data as it does not capture the variability in the data.

> *(4) Figure 3*
>
> *In the neurotranscriptome analysis of the (central) brain involving the two types of comparisons, can the authors clarify what "excluded in males" refers to? Does this imply that only genes not expressed in males were considered in the analysis? If so, what about co-expressed genes that have a specific function in female feeding behaviour?*

This is indeed correct. We reasoned that since blood feeding is exclusive to females, we should focus our analysis on genes that were specifically upregulated in them. As the reviewer points out, it is very likely that genes commonly upregulated in males and females may also promote blood feeding and we will miss out on any such candidates based on our selection criteria.

> *(5) Figure 4*
>
> *The authors state that there is more efficient knockdown in the head of unfed females; however, this is not accurate since they only get knockdown in unfed animals, and no evidence of any knockdown in fed animals (panel D). This point should be revised in the results test as well.*

Perhaps we do not understand the reviewer's point or there has been a misunderstanding. In figure 4D, we show that while there is more robust gene knockdown in unfed females, blood-fed females also showed modest but measurable knockdowns ranging from 5-40% for RYamide and 2-21% for sNPF.

> *Relatedly, blood-feeding is decreased when both neuropeptide transcripts are targeted compared to uninjected (panel C) but not compared to dsGFP injected (panel E). Why is this the case if authors showed earlier in this figure (panel B) that dsGFP does not impact blood feeding?*

We realise this concern stems from our representation of the data. Since we had earlier determined that dsGFP-injected females fed similarly to uninjected females (fig 4B), we used these controls interchangeably in subsequent experiments. To avoid confusion, we have now only used the label 'control' in figure 4 (and supplementary figure S9) and specified which control was used for each experiment in the legend.

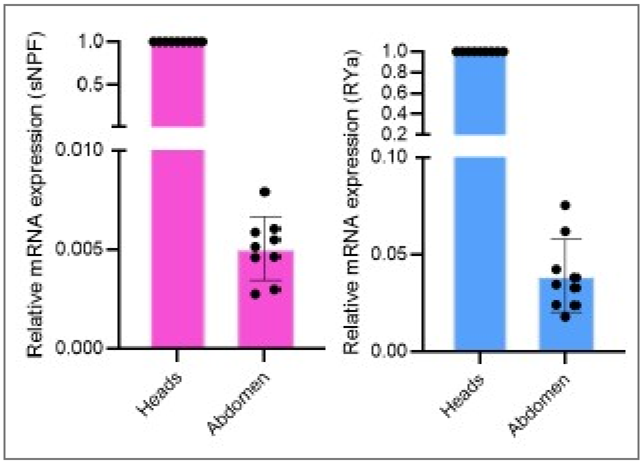
In addition to this, we wanted to clarify that fig 4C and 4E are independent experiments. 4C is the behaviour corresponding to when the neuropeptides were knocked down in both heads and abdomens. 4E is the behaviour corresponding to when the neuropeptides were knocked down in only the abdomens. We have now added a schematic in the plots to make this clearer.

> *In addition, do the uninjected and dsGFP-injected relative mRNA expression data reflect combined RYa and sNPF levels? Why is there no variation in these data,...*

In these qPCRs, we calculated relative mRNA expression using the delta-delta Ct method (see line 975). For each neuropeptide its respective control was used. For simplicity, we combined the RYa and sNPF control data into a single representation. The value of this control is invariant because this method sets the control baseline to a value of 1.

> *...and how do transcript levels of RYa and sNPF compare in the brain versus the abdomen (the presentation of data doesn't make this relationship clear).*

The reviewer is correct in pointing out that we have not clarified this relationship in our current presentation. While we have not performed absolute mRNA quantifications, we extracted relative mRNA levels from qPCR data of 96h old unmanipulated control females. We observed that both sNPF and RYa transcripts are expressed at much lower levels in the abdomens, as compared to those in the heads, as shown in Author response Image 1 below.



Author response image 1.

> *(6) As an overall comment, the figure captions are far too long and include redundant text presented in the methods and results sections.*

We thank the reviewer for flagging this and have now edited the legends to remove redundancy.

> *(7) Criteria used for identifying neuropeptides promoting blood-feeding: statement that reads "all neuropeptides, since these are known to regulate feeding behaviours". This is not accurate since not all neuropeptides govern feeding behaviors, while certainly a subset do play a role.*

We agree with the reviewer that not all neuropeptides regulate feeding behaviours. Our statement refers to the screening approach we used: in our shortlist of candidates, we chose to validate all neuropeptides.

> *(8) In the section beginning with "Two neuropeptides - sNPF and RYa - showed about 25% and 40% reduced mRNA levels...", the authors state that there was no change in blood-feeding and later state the opposite. The wording should be clarified as it is unclear.*

Thank you for pointing this out. We were referring to an unchanged proportion of the blood fed females. We have now edited the text to the following:

"Two neuropeptides - sNPF and RYa - showed about 25% and 40% reduced mRNA levels in the heads but the proportion of females that took blood meals remained unchanged". See lines 338-340.

> *(9) Just before the conclusions section, the statement that "neuropeptide receptors are often ligandpromiscuous" is unjustified. Indeed, many studies have shown in heterologous systems that high concentrations of structurally related peptides, which are not physiologically relevant, might cross-react and activate a receptor belonging to a different peptide family; however, the natural ligand is often many times more potent (in most cases, orders of magnitude) than structurally related peptides. This is certainly the case for various RYamide and sNPF receptors characterized in various insect species.*

We agree with the reviewer and apologise for the mistake. We have now removed the statement.

> *(10) Methods*
>
> *In the dsRNA-mediated gene knockdown section, the authors could more clearly describe how much dsRNA was injected per target. At the moment, the reader must carry out calculations based on the concentrations provided and the injected volume range provided later in this section.*

We have now edited the section to reflect the amount of dsRNA injected per target. Please see lines 921-931.

> *It is also unclear how tissue-specific knockdown was achieved by performing injection on different days/times. The authors need to explain/support, and justify how temporal differences in injection lead to changes in tissue-specific expression. Does the blood-brain barrier limit knockdown in the brain instead, while leaving expression in the peripheral organs susceptible?*

To achieve tissue-specific knockdowns of sNPF and RYa, we optimised both the time of injection as well as the dsRNA concentration to be injected. Injecting dsRNA into 0-10h females produced abdomen-specific knockdowns without affecting head expression, whereas injections into 96h old females resulted in knockdowns in both tissues. Head knockdowns in older females required higher dsRNA concentrations, with knockdown efficiency correlating with the amount injected. In contrast, abdominal knockdowns in younger females could be achieved even with lower dsRNA amounts, reflecting the lower baseline expression of sNPF in abdomens compared to heads and the age-dependent increase in head expression (as confirmed by qPCR). It is possible that the blood-brain barrier also limits the dsRNA entering the brain, thereby requiring higher amounts to be injected for head knockdowns.

We have now edited this section to state our methodology more clearly (see lines 932-948).

> *For example, in Figure 4, the data support that knockdown in the head/brain is only effective in unfed animals compared to uninjected animals, while there is no evidence of knockdown in the brain relative to dsGFP-injected animals. Comparatively, evidence appears to show stronger evidence of abdominal knockdown mostly for the RYa transcript (>90%) while still significantly for the sNPF transcript (>60%).*

As we explained earlier, this concern likely stems from our representation of the data. Since we had earlier determined that dsGFP-injected females fed similarly to uninjected females (fig 4B), we used these controls interchangeably in subsequent experiments. To avoid confusion, we have now only used the label 'control' in figure 4 (and supplementary figure S9) and specified which control was used for each experiment in the legend.

In addition to this, we wanted to clarify that fig 4C and 4E are independent experiments. 4C is the behaviour corresponding to when the neuropeptides were knocked down in both heads and abdomens. 4E is the behaviour corresponding to when the neuropeptides were knocked down in only the abdomen. We have now added a schematic in the plots to make this clearer.

> ***Reviewer #3 (Public review):***
>
> *Summary:*
>
> *This manuscript investigates the regulation of host-seeking behavior in Anopheles stephensi females across different life stages and mating states. Through transcriptomic profiling, the authors identify differential gene expression between "blood-hungry" and "blood-sated" states. Two neuropeptides, sNPF and RYamide, are highlighted as potential mediators of host-seeking behavior. RNAi knockdown of these peptides alters host-seeking activity, and their expression is anatomically mapped in the mosquito brain (sNPF and RYamide) and midgut (sNPF only).*

> *Strengths:*
>
> *(1) The study addresses an important question in mosquito biology, with relevance to vector control and disease transmission.*
>
> *(2) Transcriptomic profiling is used to uncover gene expression changes linked to behavioral states.*
>
> *(3) The identification of sNPF and RYamide as candidate regulators provides a clear focus for downstream mechanistic work.*
>
> *(4) RNAi experiments demonstrate that these neuropeptides are necessary for normal host-seeking behavior.*
>
> *(5) Anatomical localization of neuropeptide expression adds depth to the functional findings.*
>
> *Weaknesses:*
>
> *(1) The title implies that the neuropeptides promote host-seeking, but sufficiency is not demonstrated (for example, with peptide injection or overexpression experiments).*

Demonstrating sufficiency would require injecting sNPF peptide or its agonist. To date, no small-molecule agonists (or antagonists) that selectively mimic sNPF or RYa neuropeptides have been identified in insects. An NPY analogue, TM30335, has been reported to activate the Aedes aegypti NPY-like receptor 7 (NPYLR7; Duvall et al., 2019), which is also activated by sNPF peptides at higher doses (Liesch et al., 2013). Unfortunately, the compound is no longer available because its manufacturer, 7TM Pharma, has ceased operations. Synthesising the peptides is a possibility that we will explore in the future.

> *(2) The proposed model regarding central versus peripheral (gut) peptide action is inconsistently presented and lacks strong experimental support.*

The best way to address this would be to conduct tissue-specific manipulations, the tools for which are not available in this species. Our approach to achieve head+abdomen and abdomen only knockdown was the closest we could get to achieving tissue specificity and allowed us to confirm that knockdown in the head was necessary for the phenotype. However, as the reviewer points out, this did not allow us to rule out any involvement of the abdomen. This point has been addressed in lines 364-371.

> *(3) Some conclusions appear premature based on the current data and would benefit from additional functional validation.*

The most definitive way of demonstrating necessity of sNPF and RYa in blood feeding would be to generate mutant lines. While we are pursuing this line of experiments, they lie beyond the scope of a revision. In its absence, we relied on the knockdown of the genes using dsRNA. We would like to posit that despite only partial knockdown, mosquitoes do display defects in blood-feeding behaviour, without affecting sugar-feeding. We think this reflects the importance of sNPF in promoting blood feeding.

> ***Recommendations for the authors:***
>
> ***Reviewer #1 (Recommendations for the authors):***
>
> *Overall, I found this manuscript to be well-prepared, visually the figures are great and clearly were carefully thought out and curated, and the research is impactful. It was a wonderful read from start to finish. I have the following recommendations:*

Thank you very much, we are very pleased to hear that you enjoyed reading our manuscript!

> *(1) For future manuscripts, it would make things significantly easier on the reviewer side to submit a format that uses line numbers.*

We sincerely apologise for the oversight. We have now incorporated line numbers in the revised manuscript.

> *(2) There are a few statements in the text that I think may need clarification or might be outside the bounds of what was actually studied here. For example, in the introduction "However, mating is dispensable in Anophelines even under conditions of nutritional satiety". I am uncertain what is meant by this statement - please clarify.*

We apologise for the lack of clarity in the statement and have now deleted it since we felt it was not necessary.

> *(3) Typo/Grammatical minutiae:*
>
> *(a) A small idiosyncrasy of using hyphens in compound words should also be fixed throughout. Typically, you don't hyphenate if the words are being used as a noun, as in the case: e.g. "Age affects blood feeding.". However, you would hyphenate if the two words are used as a compound adjective "Age affects blood-feeding behavior". This may not be an all-inclusive list, but here are some examples where hyphens need to either be removed or added. Some examples:*
>
> *"Nutritional state also influences other internal state outputs on blood-feeding": blood-feeding -> blood feeding*
>
> *"... the modulation of blood-feeding": blood-feeding -> blood feeding*
>
> *"For example, whether virgin females take blood-meals...": blood-meals -> blood meals*
>
> *".... how internal and external cues shape meal-choice"-> meal choice*
>
> *"blood-meal" is often used throughout the text, but is correctly "blood meal" in the figures.*
>
> *There are many more examples throughout.*

We apologise for these errors and appreciate the reviewer's keen eye. We have now fixed them throughout the manuscript.

> *(b) Figure 1 Caption has a typo: "co-housed males were accessed for sugar-feeding" should be "co-housed males were assessed for sugar feeding"*

We apologise for the typo and thank the reviewer for spotting it. We have now corrected this.

> *(c) It would be helpful in some other figure captions to more clearly label which statement is relevant to which part of the text. For example, in Figure 4's caption.*
>
> *"C,D. Blood-feeding and sugar-feeding behaviour of females when both RYa and sNPF are knocked down in the head (C). Relative mRNA expressions of RYa and sNPF in the heads of dsRYa+dssNPF - injected blood-fed and unfed females, as compared to that in uninjected females, analysed via qPCR (D)."*
>
> *I found re-referencing C and D at the end of their statements makes it look as thought C precedes the "Relative mRNA expression" and on a first read through, I thought the figure captions were backwards. I'd recommend reformatting here and throughout consistently to only have the figure letter precede its relevant caption information, e.g.:*

> *"C. Blood-feeding and sugar-feeding behaviour of females when both RYa and sNPF are knocked down in the head. D. Relative mRNA expressions of RYa and sNPF in the heads of dsRYa+dssNPF - injected bloodfed and unfed females, as compared to that in uninjected females, analysed via qPCR."*

We have now edited the legends as suggested.

> ***Reviewer #2 (Recommendations for the authors):***
>
> *Separately from the clarifications and limitations listed above, the authors could strengthen their study and the conclusions drawn if they could rescue the behavioural phenotype observed following knockdown of sNPF and RYamide. This could be achieved by injection of either sNPF or RYa peptide independently or combined following knockdown to validate the role of these peptides in promoting blood-feeding in An. stephensi. Additionally, the apparent (but unclear) regionalized (or tissue-specific) knockdown of sNPF and RYamide transcripts could be visualized and verified by implementing HCR in situ hyb in knockdown animals (or immunohistochemistry using antibodies specific for these two neuropeptides).*

In a follow up of this work, we are generating mutants and peptides for these candidates and are planning to conduct exactly the experiments the reviewer suggests.

> ***Reviewer #3 (Recommendations for the authors):***
>
> *The loss-of-function data suggest necessity but not sufficiency. Synthetic peptide injection in non-hostseeking (blood-fed mated or juvenile) mosquitoes would provide direct evidence for peptide-induced behavioral activation. The lack of these experiments weakens the central claim of the paper that these neuropeptides directly promote blood feeding.*

As noted above, we plan to synthesise the peptide to test rescue in a mutant background and sufficiency.

> *Some of the claims about knockdown efficiency and interpretation are conflicting; the authors dismiss Hairy and Prp as candidates due to 30-35% knockdown, yet base major conclusions on sNPF and RYamide knockdowns with comparable efficiencies (25-40%). This inconsistency should be addressed, or the justification for different thresholds should be clearly stated.*

We have not defined any specific knockdown efficacy thresholds in the manuscript, as these can vary considerably between genes, and in some cases, even modest reductions can be sufficient to produce detectable phenotypes. For example, knockdown efficiencies of even as low as about 25% - 40% gave us observable phenotypes for sNPF and RYa RNAi (Figure S9B-G).

No such phenotypes were observed for Hairy (30%) or Prp (35%) knockdowns. Either these genes are not involved in blood feeding, or the knockdown was not sufficient for these specific genes to induce phenotypes. We cannot distinguish between these scenarios.

> *The observation that knockdown animals take smaller blood meals is interesting and could reflect a downstream effect of altered host-seeking or an independent physiological change. The relationship between meal size and host-seeking behavior should be clarified.*

We agree with the reviewer that the reduced meal size observed in sNPF and RYa knockdown animals could result from their inability to seek a host or due to an independent effect on blood meal intake. Unfortunately, we did not measure host-seeking in these animals. We plan to distinguish between these possibilities using mutants in future work.

> *Several figures are difficult to interpret due to cluttered labeling and poorly distinguishable color schemes. Simplifying these and improving contrast (especially for co-housed vs. virgin conditions) would enhance readability.*

We regret that the reviewer found the figures difficult to follow. We have now revised our annotations throughout the manuscript for enhanced readability. For example, "D1^{B}" is now "D1^{PBM}" (post-bloodmeal) and "D1^{O}" is now "D1^{PO}" (post-oviposition). Wherever mated females were used, we have now appended "(m)" to the annotations and consistently depicted these females with striped abdomens in all the schematics. We believe these changes will improve clarity and readability.

> *The manuscript does not clearly justify the use of whole-brain RNA sequencing to identify peptides involved in metabolic or peripheral processes. Given that anticipatory feeding signals are often peripheral, the logic for brain transcriptomics should be explained.*

The reviewer is correct in pointing out that feeding signals could also emerge from peripheral tissues. Signals from these tissues – in response to both changing nutritional and reproductive states – are then integrated by the central brain to modulate feeding choices. For example, in Drosophila, increased protein intake is mediated by central brain circuitry including those in the SEZ and central complex (Munch et al., 2022; Liu et al., 2017; Goldschmidt et al., 202ti). In the context of mating, male-derived sex peptide further increases protein feeding by acting on a dedicated central brain circuitry (Walker et al., 2015). We, therefore focused on the central brain for our studies.

> *The proposed model suggests brain-derived peptides initiate feeding, while gut peptides provide feedback. However, gut-specific knockdowns had no effect, undermining this hypothesis. Conversely, the authors also suggest abdominal involvement based on RNAi results. These contradictions need to be resolved into a consistent model.*

We thank the reviewer for raising this point and recognise their concern. Our reasons for invoking an involvement of the gut were two-fold:

(1) We find increased sNPF transcript expression in the entero-endocrine cells of the midgut in blood-hungry females, which returns to baseline after a blood-meal (Fig. 4L, M).

(2) While the abdomen-only knockdowns did not affect blood feeding, every effective head knockdown that affected blood feeding also abolished abdominal transcript levels (Fig. S9C, F). (Achieving a head-only reduction proved impossible because (i) systemic dsRNA delivery inevitably reaches the abdomen and (ii) abdominal expression of both peptides is low, leaving little dynamic range for selective manipulation.) Consequently, we can only conclude the following: 1) that brain expression is required for the behaviour, 2) that we cannot exclude a contributory role for gut-derived sNPF. We have discussed this in lines 364-371.

> *The identification of candidate receptors is promising, but the manuscript would be significantly strengthened by testing whether receptor knockdowns phenocopy peptide knockdowns. Without this, it is difficult to conclude that the identified receptors mediate the behavioral effects.*

We agree that functional validation of the receptors would strengthen the evidence for sNPF and *RYa*-mediated control of blood feeding in An. stephensi. We selected these receptors based on sequence homology. A possibility remains that sNPF neuropeptides activate more than one receptor, each modulating a distinct circuit, as shown in the case of Drosophila Tachykinin (*https://pmc.ncbi.nlm.nih.gov/articles/PMC10184743/*). This will mean a systematic characterisation and knockdown of each of them to confirm their role. We are planning these experiments in the future.

> *The authors compared the percentage changes in sugar-fed and blood-fed animals under sugar-sated or sugar-starved conditions. Figure 1F should reflect what was discussed in the results.*

Perhaps this concern stems from our representation of the data in figure 1F? We have now edited the xaxis and revised its label from “choice of food” to “choice made” to better reflect what food the mosquitoes chose to take.

For clarity, we have now also plotted the same data as stacked graphs at the bottom of Fig. 1F, which clearly shows the proportion of mosquitoes fed on each particular choice. We avoid the stacked graph as the sole representation of this data because it does not capture the variability in the data.

> *Minor issues:*
>
> *(1) The authors used mosquitoes with belly stripes to indicate mated females. To be consistent, the post-oviposition females should also have belly stripes.*

We thank the reviewer for pointing this out. We have now edited all the figures as suggested.

> *(2) In the first paragraph on the right column of the second page, the authors state, "Since females took blood-meals regardless of their prior sugar-feeding status and only sugar-feeding was selectively suppressed by prior sugar access." Just because the well-fed animals ate less than the starved animals does not mean their feeding behavior was suppressed.*

Perhaps there has been a misunderstanding in the experimental setup of figure 1F, probably stemming from our data representation. The experiment is a choice assay in which sugar-starved or sugar-sated females, co-housed with males, were provided simultaneous access to both blood and sugar, and were assessed for the choice made (indicated on the x-axis): both blood and sugar, blood only, sugar only, or neither. We scored females only for the presence or absence of each meal type (blood or sugar) and did not quantify the amount consumed.

> *(3) The figure legend for Figure 1A and the naming convention for different experimental groups are difficult to follow. A simplified or consistently abbreviated scheme would help readers navigate the figures and text.*

We regret that the reviewer found the figure difficult to follow. We have now revised our annotations throughout the manuscript for enhanced readability. For example, “$D1^{B}$” is now “$D1^{PBM}$” (post-bloodmeal) and “$D1^{O}$” is now “$D1^{PO}$” (post-oviposition).

> *(4) In the last paragraph of the Y-maze olfactory assay for host-seeking behaviour in An. stephensi in Methods, the authors state, "When testing blood-fed females, aged-matched sugar-fed females (bloodhungry) were included as positive controls where ever possible, with satisfactory results." The authors should explicitly describe what the criteria are for "satisfactory results".*

We apologise for the lack of clarity. We have now edited the statement to read:

“When testing blood-fed females, age-matched sugar-fed females (blood-hungry) were included wherever possible as positive controls. These females consistently showed attraction to host cues, as expected.” See lines 786-790.

> *(5) In the first paragraph of the dsRNA-mediated gene knockdown section in Methods, dsRNA against GFP is used as a negative control for the injection itself, but not for the*

> *potential off-target effect.*

We agree with the reviewer that dsGFP injections act as controls only for injection-related behavioural changes, and not for off-target effects of RNAi. We have now corrected the statement. See lines 919-920.

To control for off-target effects, we could have designed multiple dsRNAs targeting different parts of a given gene. We regret not including these controls for potential off-target effects of dsRNAs injected.

> *(6) References numbers 48, 89, and 90 are not complete citations.*

We thank the reviewer for spotting these. We have now corrected these citations.

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